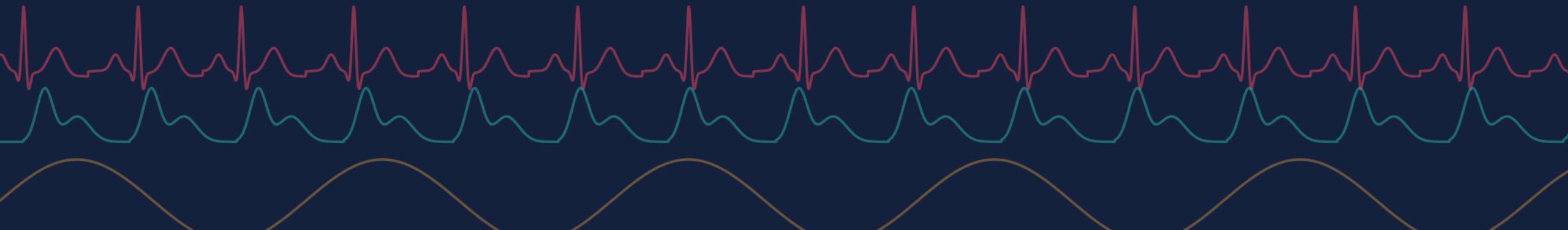
HRV Instrumentation

Sensors, signals and artifacts: measuring heart rate variability with PPG, ECG and respiration sensors

Biofeedback Technologies (BFTa)
Prof. Aamir Malik · Faculty of Information Technology, Brno University of Technology

Topic coverage adapted from Shaffer, Khazan & Moss, BCIA HRV Biofeedback Didactic Workshop: “HRV Instrumentation” (2024). Text, figures and analyses in this deck are original; signal examples use simulated data and open PhysioNet recordings; photos are openly licensed and credited.

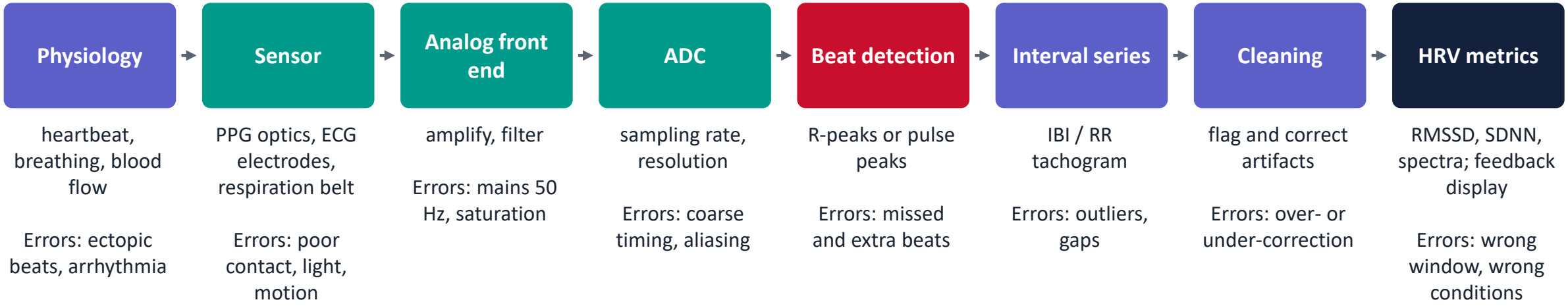


What you will be able to do after this lecture

1. Explain how **PPG, ECG and respiration sensors** turn physiology into electrical signals, and what limits each of them.
2. **Choose and place** the right sensor for an HRV application.
3. **Recognise the main artifacts** in raw signals and know how to prevent or correct them.
4. Describe the **processing chain** from raw signal to HRV metrics: filtering, sampling, beat detection, artifact correction.
5. Judge whether a recording is **clean enough to analyse**.

- **HRV primer**
- **Part 1 · PPG (blood volume pulse)**
- **Part 2 · ECG**
- **Part 3 · Respiration**
- **Beyond the lab and in practice**

Every HRV number is the end of a measurement chain, and every link can add error



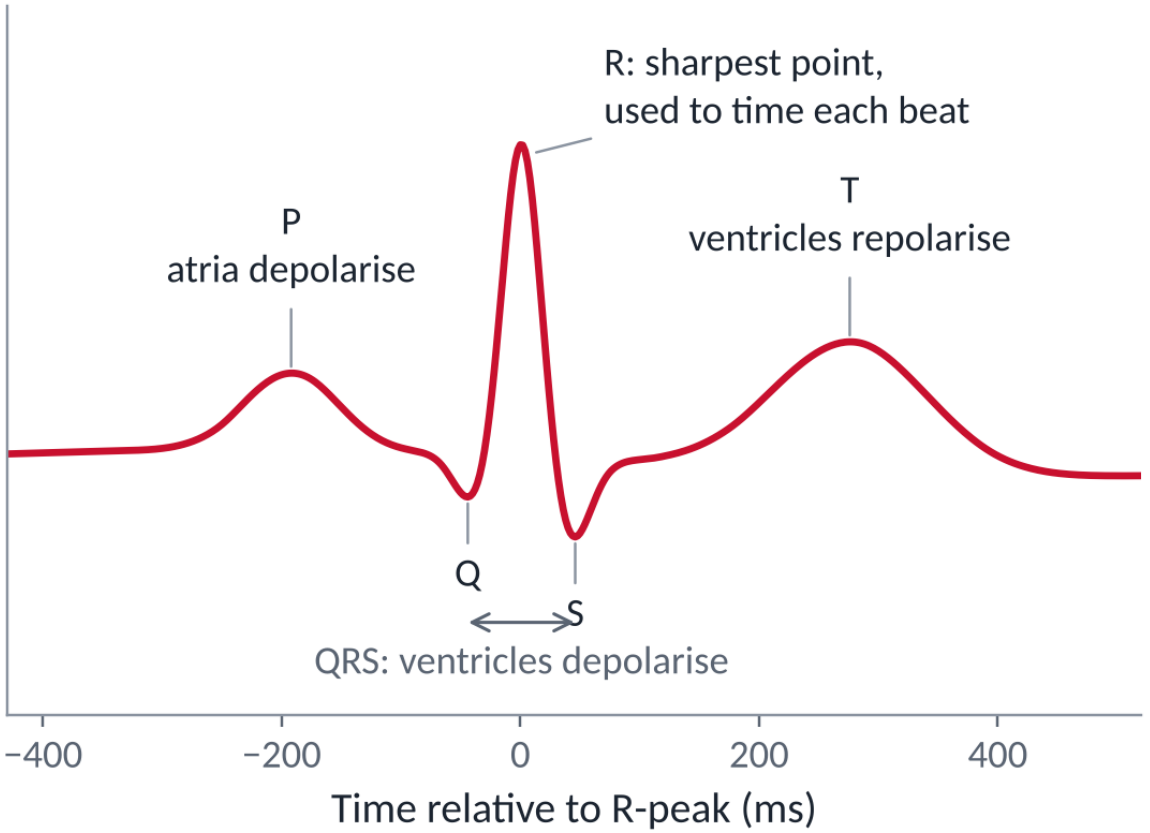
Garbage in, garbage out. No algorithm can fully rescue a bad recording. Most of this lecture is about getting clean signals into the chain and recognising when you have not.

Peper, E., et al. (2010). Garbage in; garbage out: Identify blood volume pulse (BVP) artifacts before analyzing and interpreting BVP, BVP amplitude, and HR/RSA data. Biofeedback, 38(1), 19–23. doi:10.5298/1081-5937-38.1.19

Quigley, K. S., et al. (2024). Publication guidelines for human heart rate and heart rate variability studies in psychophysiology, Part 1. Psychophysiology, 61(9), e14604. doi:10.1111/psyp.14604

The heart sets its own rhythm; the autonomic nervous system speeds it up or slows it down

- Each beat starts in the **sinoatrial (SA) node**, the natural pacemaker in the right atrium.
- The impulse spreads over the atria (**P wave**), is delayed in the **atrioventricular (AV) node**, then travels down the His-Purkinje fibres so both ventricles contract together (**QRS**). Ventricles then recover (**T wave**).
- Left alone, the SA node fires at about **100 beats/min** in young adults (lower with age). At rest, the **vagus nerve** (parasympathetic) holds it down to roughly 60 to 80 beats/min.
- **Vagal** effects act within one or two beats; **sympathetic** effects (faster, stronger beats) build up over seconds.
- So fast, beat-to-beat changes in heart rate mostly reflect **vagal** activity: the basis of many HRV measures.



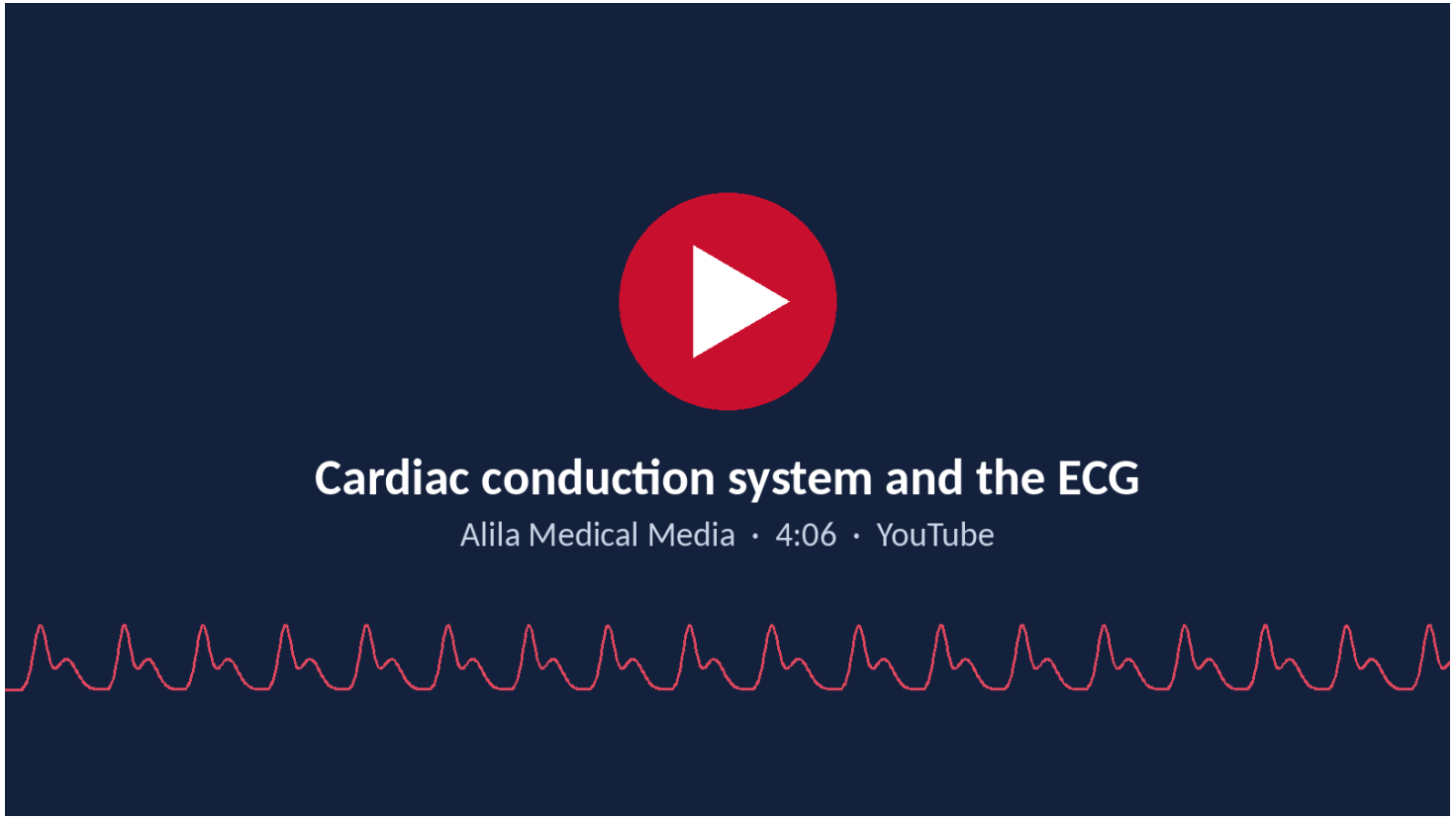
One ECG beat (simulated) with the cardiac event behind each wave

Shaffer, F., & Ginsberg, J. P. (2017). An overview of heart rate variability metrics and norms. Front Public Health, 5, 258. doi:10.3389/fpubh.2017.00258

Jose, A. D., & Collison, D. (1970). The normal range and determinants of the intrinsic heart rate in man. Cardiovasc Res, 4(2), 160–167. doi:10.1093/cvr/4.2.160

McSharry, P. E., et al. (2003). A dynamical model for generating synthetic electrocardiogram signals. IEEE Trans Biomed Eng, 50(3), 289–294. doi:10.1109/TBME.2003.808805

Video: how the electrical impulse travels through the heart and creates the ECG (4 min)



If the video does not play here, open it on YouTube

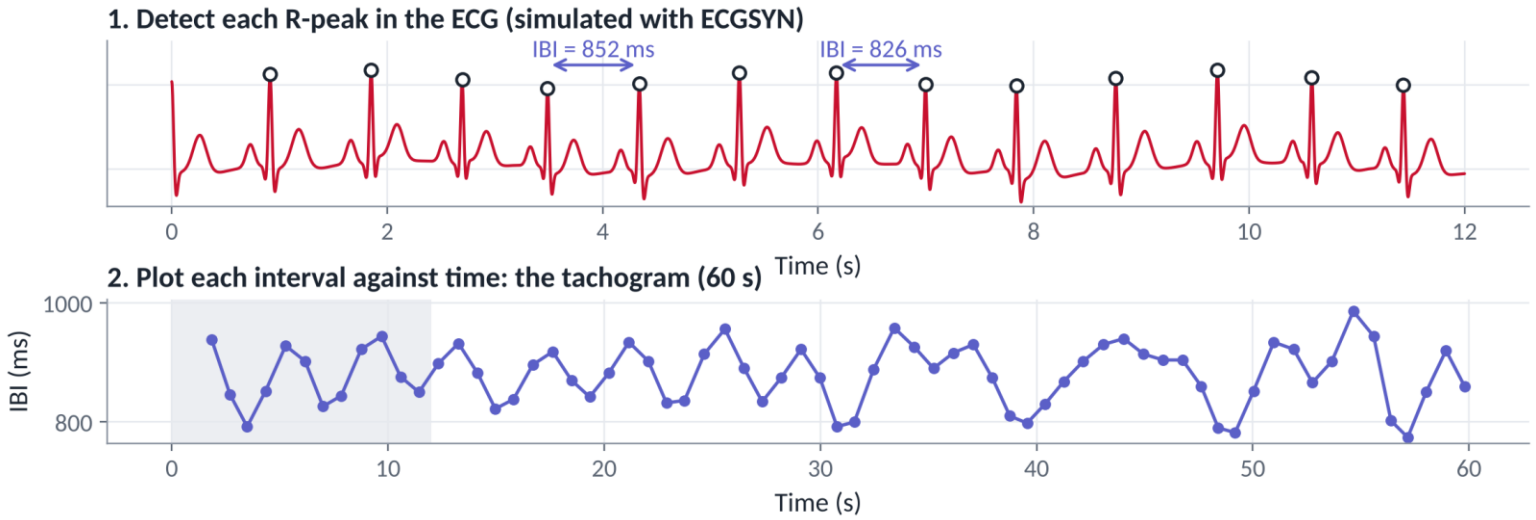
While you watch, look for:

- Where each beat starts (SA node).
- Why there is a short pause at the AV node.
- Which ECG wave matches which event (P, QRS, T).
- Why the shape of the ECG depends on where the electrodes are placed (we use this in Part 2).

Video: Alila Medical Media (2024). Cardiac conduction system and understanding ECG/EKG, updated 2024. YouTube. https://www.youtube.com/watch?v=ScZVGHDisa0

HRV analysis starts by turning a waveform into a list of beat times

- Pick **one timing point per beat**: the R-peak in ECG; the systolic peak or the pulse foot in PPG.
- **Interbeat interval (IBI)** = time between successive beats (ms). With ECG it is called the **RR interval**; **NN** means normal-to-normal, i.e. after artifacts and ectopic beats are removed.
- **Heart rate** (beats/min) = 60 000 / IBI (ms). Example: 800 ms gives 75 beats/min.
- The **tachogram** (IBI against time) is the input to every HRV metric. Note: it has one value per beat, so it is **unevenly sampled**.

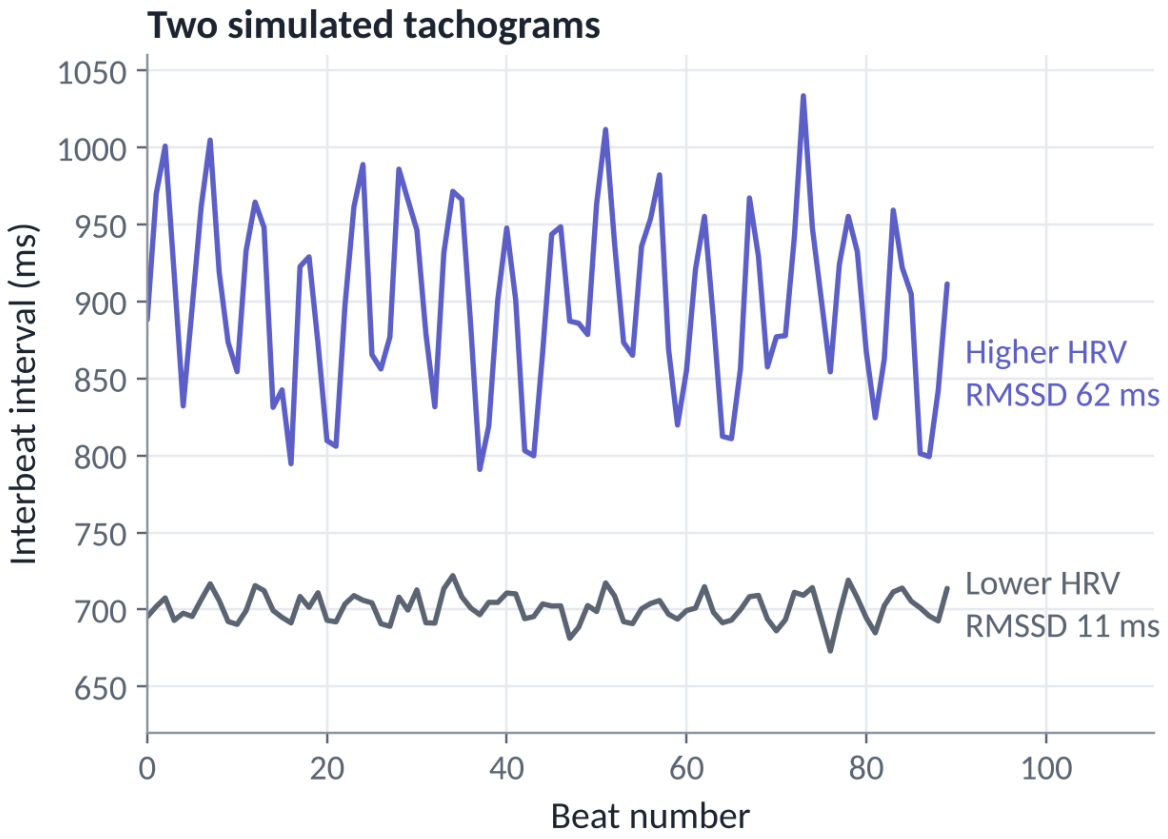


Task Force of the ESC and NASPE (1996). Heart rate variability: Standards of measurement, physiological interpretation and clinical use. Circulation, 93(5), 1043–1065. doi:10.1161/01.CIR.93.5.1043

McSharry, P. E., et al. (2003). A dynamical model for generating synthetic electrocardiogram signals. IEEE Trans Biomed Eng, 50(3), 289–294. doi:10.1109/TBME.2003.808805

HRV measures how much the time between heartbeats changes: a window on autonomic regulation

- A healthy heart is **not a metronome**: intervals change continuously with breathing, blood-pressure regulation, posture, emotion and effort.
- Higher resting HRV generally indicates stronger **vagal (parasympathetic) influence** and more flexible regulation; low HRV is associated with ageing, chronic stress and poorer cardiovascular outcomes.
- HRV depends strongly on **age, heart rate, breathing and recording conditions**: always compare like with like.
- **HRV biofeedback** trains people to increase HRV, mainly through slow paced breathing. The sensors in this lecture provide the real-time feedback.



Shaffer, F., & Ginsberg, J. P. (2017). An overview of heart rate variability metrics and norms. Front Public Health, 5, 258. doi:10.3389/fpubh.2017.00258

Laborde, S., et al. (2017). Heart rate variability and cardiac vagal tone in psychophysiological research: Recommendations for experiment planning, data analysis, and data reporting. Front Psychol, 8, 213. doi:10.3389/fpsyg.2017.00213

Lehrer, P. M., & Gevirtz, R. (2014). Heart rate variability biofeedback: How and why does it work? Front Psychol, 5, 756. doi:10.3389/fpsyg.2014.00756

A few metrics cover most HRV work; all are computed from the interval series

Domain	Metric	How it is computed	Mainly reflects
Time	**SDNN** (ms)	Standard deviation of all NN intervals	Overall variability, all sources
Time	**RMSSD** (ms)	Root mean square of successive differences (formula below)	Fast, beat-to-beat (vagal) changes
Time	**pNN50** (%)	Percentage of successive differences larger than 50 ms	Vagal; closely tracks RMSSD
Time	**HR Max – HR Min** (beats/min)	Per breath: highest minus lowest HR; averaged over breaths	Respiratory sinus arrhythmia; needs a breathing signal
Frequency	**HF power** 0.15–0.40 Hz	Spectrum of the resampled tachogram	Breathing-related, vagal
Frequency	**LF power** 0.04–0.15 Hz	Same spectrum, lower band	Blood-pressure (baroreflex) rhythm; slow breathing lands here
Frequency	**VLF** 0.0033–0.04 Hz	Same spectrum, very low band	Slow regulation; needs long recordings
Nonlinear	**SD1 / SD2**	Spread of the Poincaré plot (each IBI against the next)	SD1 ≈ RMSSD / √2: short-term; SD2: longer-term

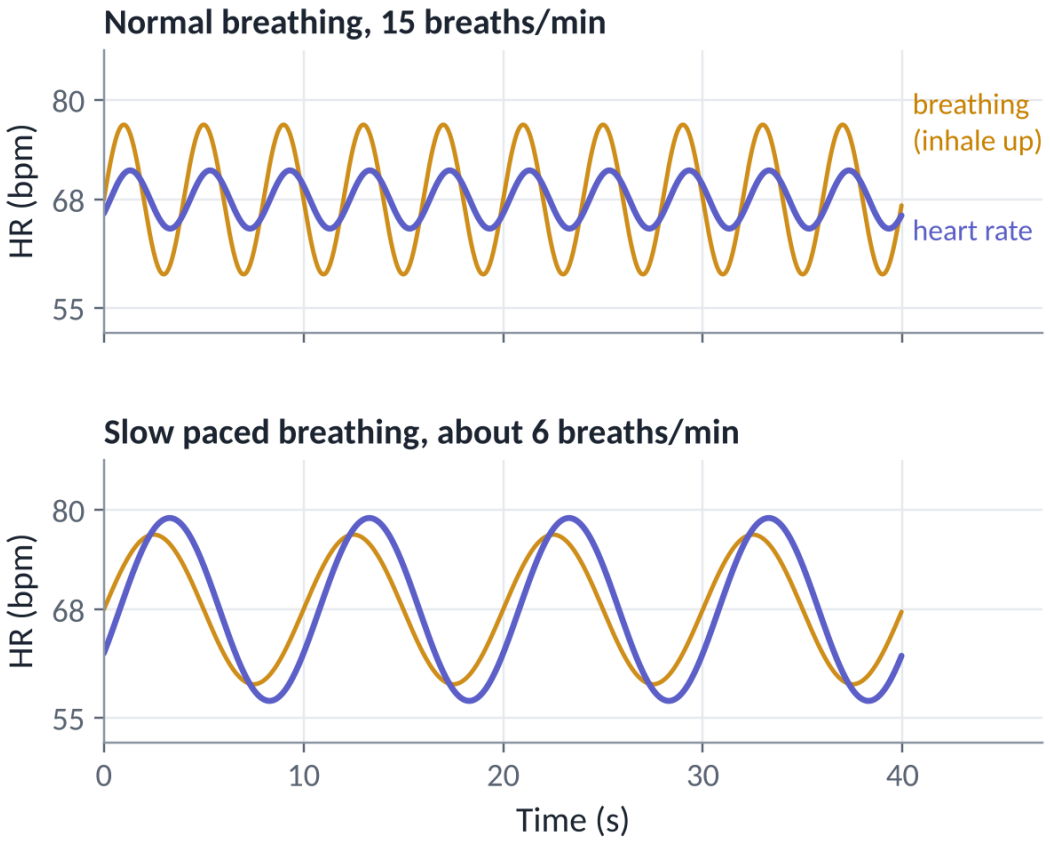
$$\mathrm{RMSSD} = \sqrt{\left((1 / (N - 1)) \cdot \Sigma \left(\mathrm{IBI}_{i+1} - \mathrm{IBI}_i\right)^2 \right)},\ \text{summed over } i = 1 \ldots N - 1,\ \text{with } N \text{ intervals}$$

Task Force of the ESC and NASPE (1996). Heart rate variability: Standards of measurement, physiological interpretation and clinical use. Circulation, 93(5), 1043–1065. doi:10.1161/01.CIR.93.5.1043

Shaffer, F., & Ginsberg, J. P. (2017). An overview of heart rate variability metrics and norms. Front Public Health, 5, 258. doi:10.3389/fpubh.2017.00258

Breathing modulates heart rate beat by beat: respiratory sinus arrhythmia (RSA)

- **Inhalation** briefly reduces vagal outflow, so heart rate rises; **exhalation** restores it, so heart rate falls.
- RSA is the main source of fast (HF) HRV at normal breathing rates of about 9 to 24 breaths/min (0.15–0.4 Hz).
- Breathing slowly at about **6 breaths/min (0.1 Hz)** lines RSA up with the blood-pressure (baroreflex) rhythm and produces very large heart-rate swings: **resonance frequency breathing**, the core of HRV biofeedback.
- This is why HRV work records **breathing as well as heartbeats** (Part 3).



Simulated illustration: same person, two breathing rates

Lehrer, P. M., et al. (2000). Resonant frequency biofeedback training to increase cardiac variability: Rationale and manual for training. Appl Psychophysiol Biofeedback, 25(3), 177–191. doi:10.1023/A:1009554825745

Lehrer, P. M., & Gevirtz, R. (2014). Heart rate variability biofeedback: How and why does it work? Front Psychol, 5, 756. doi:10.3389/fpsyg.2014.00756

Shaffer, F., & Ginsberg, J. P. (2017). An overview of heart rate variability metrics and norms. Front Public Health, 5, 258. doi:10.3389/fpubh.2017.00258

Three sensors cover HRV work: PPG and ECG for heartbeats, a respiration belt for breathing

	PPG (blood volume pulse)	ECG (electrocardiogram)	Respiration belt
Senses	Blood volume changes in skin (optical)	Electrical activity of the heart, about 1 mV at the skin	Expansion of chest or abdomen
Principle	LED light + photodiode	Electrodes + differential amplifier	Stretch changes resistance, inductance or charge
Beat marker	Systolic peak or pulse foot	R-peak	None (gives breathing rate, depth, pattern)
Timing accuracy	Good at rest; pulse travel time varies	Best: sharp R-peak; reference method	Not applicable
Set-up	Seconds; no skin preparation	Skin preparation, electrodes, sometimes clothing moved	Seconds
Main weaknesses	Movement, cold hands, room light, pressure	Mains, muscle (EMG), contact, polarity	Slipping, loose band, clipping
Typical devices	Finger and ear clips, smartwatches	Chest straps, lab amplifiers, Holter	Biofeedback belts, sleep-lab belts

Allen, J. (2007). Photoplethysmography and its application in clinical physiological measurement. Physiol Meas, 28(3), R1–R39. doi:10.1088/0967-3334/28/3/R01

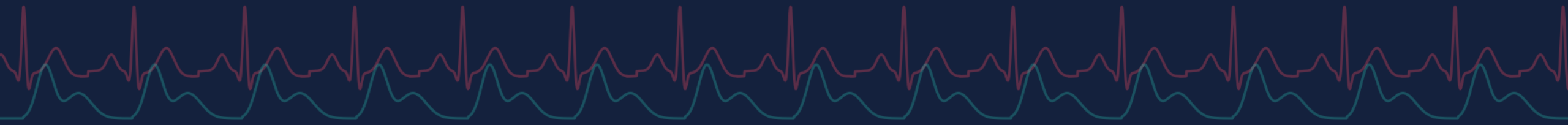
Quigley, K. S., et al. (2024). Publication guidelines for human heart rate and heart rate variability studies in psychophysiology, Part 1. Psychophysiology, 61(9), e14604. doi:10.1111/psyp.14604

Massaroni, C., et al. (2019). Contact-based methods for measuring respiratory rate. Sensors, 19(4), 908. doi:10.3390/s19040908

PART 1

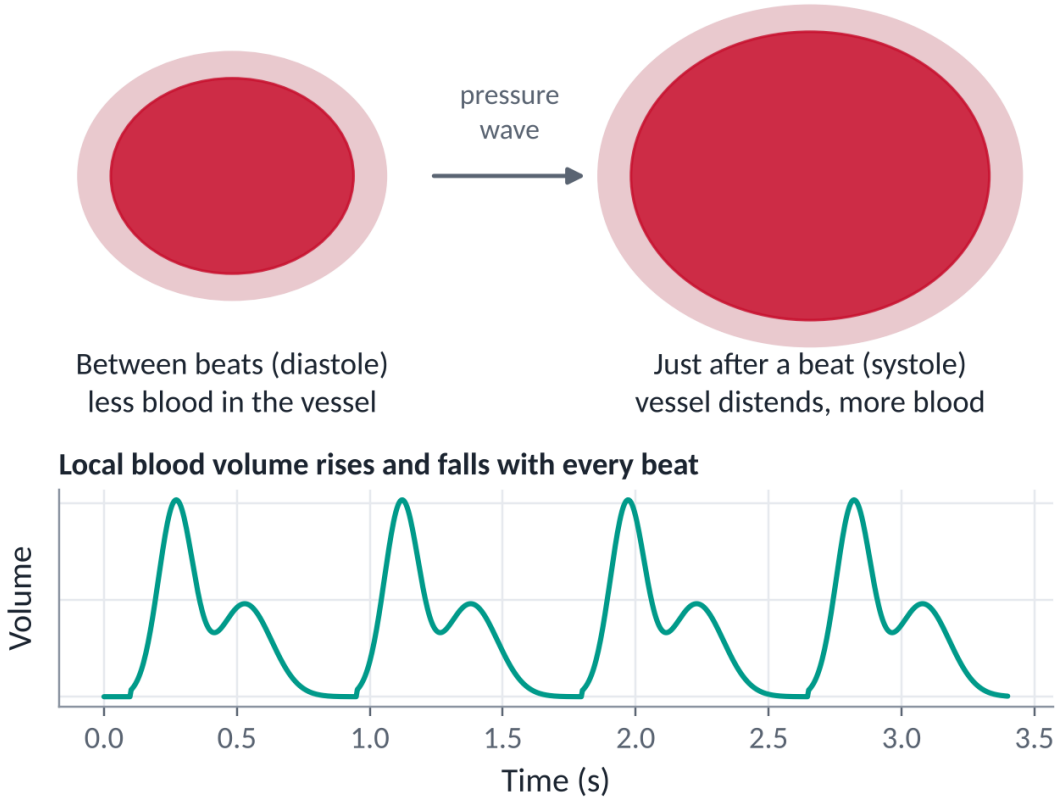
Blood volume pulse (PPG)

How an optical sensor sees each heartbeat, where to put it, how the signal is processed, and what goes wrong



Blood volume pulse (BVP) is the small rise in blood volume in the skin with every heartbeat

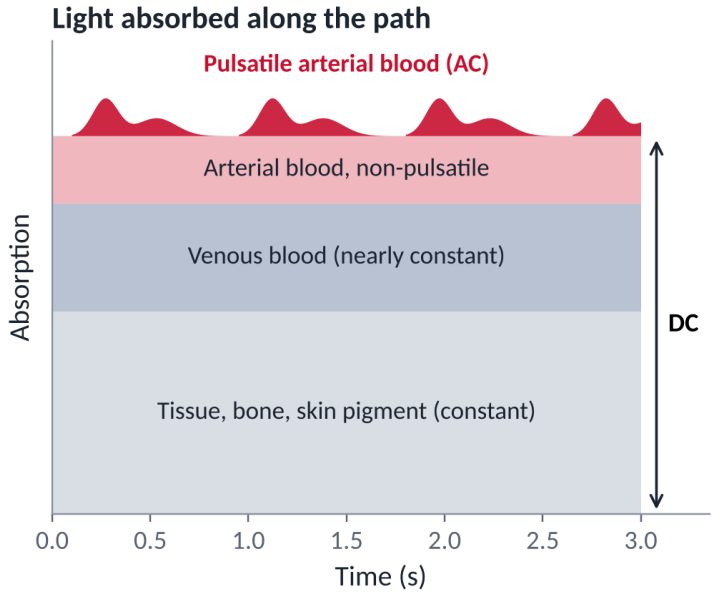
- Each heartbeat sends a **pressure wave** through the arteries; small arteries and **arterioles** in the skin briefly widen and hold more blood.
- Arterioles are small **muscular** arteries. Their tone (narrowing or widening) is set by sympathetic nerves and by temperature, so it changes the size of each pulse.
- The sensor is a **photoplethysmograph (PPG)**: photo (light) + plethysmo (volume) + graph (recording).
- In this course **BVP** and **PPG signal** mean the same thing. Pulse rate from PPG is used as a stand-in for heart rate.

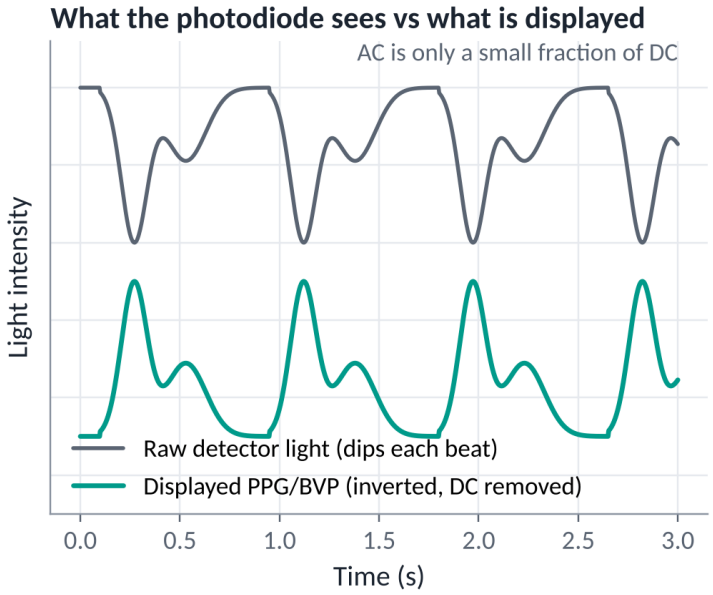


Allen, J. (2007). Photoplethysmography and its application in clinical physiological measurement. Physiol Meas, 28(3), R1–R39. doi:10.1088/0967-3334/28/3/R01

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

PPG measures how much light the tissue absorbs: more blood means less light at the detector





- An LED shines into the tissue; haemoglobin absorbs part of the light. More absorber in the path means more absorption (**Beer-Lambert law**).
- Most absorption is constant (**DC**): skin, bone, venous blood. Only a small **pulsatile (AC)** part follows the heartbeat.
- With each beat **more blood absorbs more light, so the photodiode receives less**. Devices invert and high-pass filter the signal so the displayed BVP rises with each beat.

Common misconception: blood does not “reflect more light” as volume rises. It absorbs more.

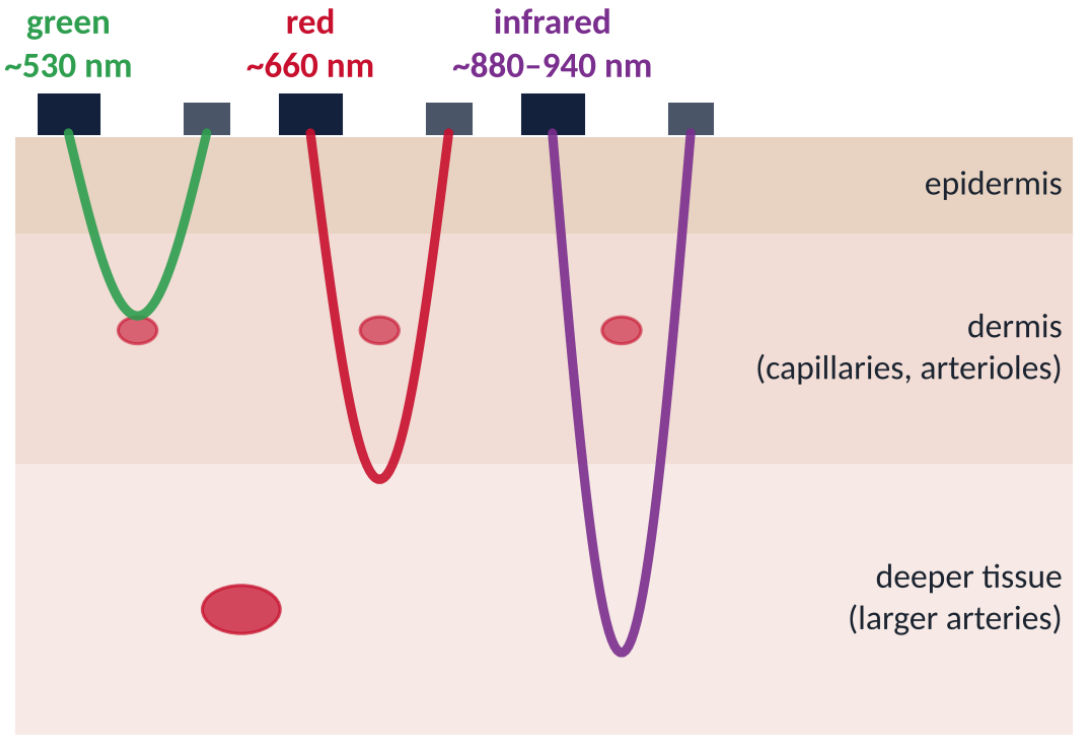
Allen, J. (2007). Photoplethysmography and its application in clinical physiological measurement. Physiol Meas, 28(3), R1–R39. doi:10.1088/0967-3334/28/3/R01

Tamura, T., et al. (2014). Wearable photoplethysmographic sensors: Past and present. Electronics, 3(2), 282–302. doi:10.3390/electronics3020282

The wavelength decides how deep the light goes and what the sensor is good for

Light	Wavelength	Reaches	Typical use
Green	≈ 530 nm	Shallow: capillaries, small arterioles	Wrist wearables: large pulsatile signal, less motion-sensitive
Red	≈ 660 nm	Deeper	Pulse oximetry, together with infrared
Infrared	≈ 880–940 nm	Deepest	Finger and ear clips, biofeedback PPG sensors, pulse oximetry

- Haemoglobin absorbs green strongly, giving a large AC signal near the surface; melanin in the skin also absorbs visible light, which weakens the signal in darker skin (slide 34).
- Red plus infrared allow **SpO_2** estimation, because oxygenated and deoxygenated haemoglobin absorb them differently.



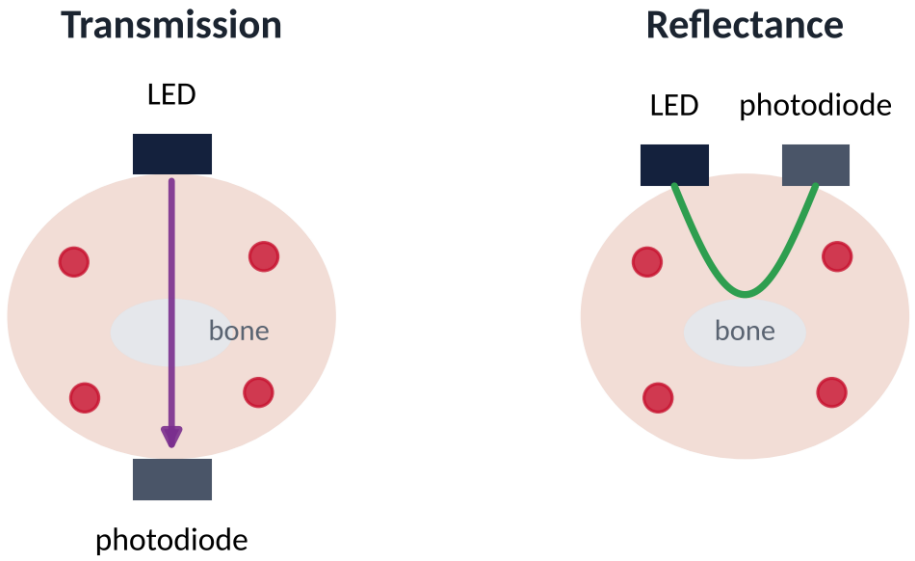
Dark block = LED, grey block = photodiode; curves show typical photon paths (illustrative)

Tamura, T., et al. (2014). Wearable photoplethysmographic sensors: Past and present. Electronics, 3(2), 282–302. doi:10.3390/electronics3020282

Charlton, P. H., et al. (2022). Wearable photoplethysmography for cardiovascular monitoring. Proc IEEE, 110(3), 355–381. doi:10.1109/JPROC.2022.3149785

Fallow, B. A., et al. (2013). Influence of skin type and wavelength on light wave reflectance. J Clin Monit Comput, 27(3), 313–317. doi:10.1007/s10877-013-9436-7

Transmission sensors shine through tissue; reflectance sensors read light scattered back



	Transmission	Reflectance
LED and detector	Opposite sides of the tissue	Same side, next to each other
Possible sites	Thin parts: fingertip, earlobe	Almost anywhere: wrist, forehead, forearm, ankle
Signal quality	Usually stronger and cleaner	Weaker; more affected by room light, skin tone, tissue thickness
Typical devices	Finger and ear clips	Smartwatches, forehead sensors

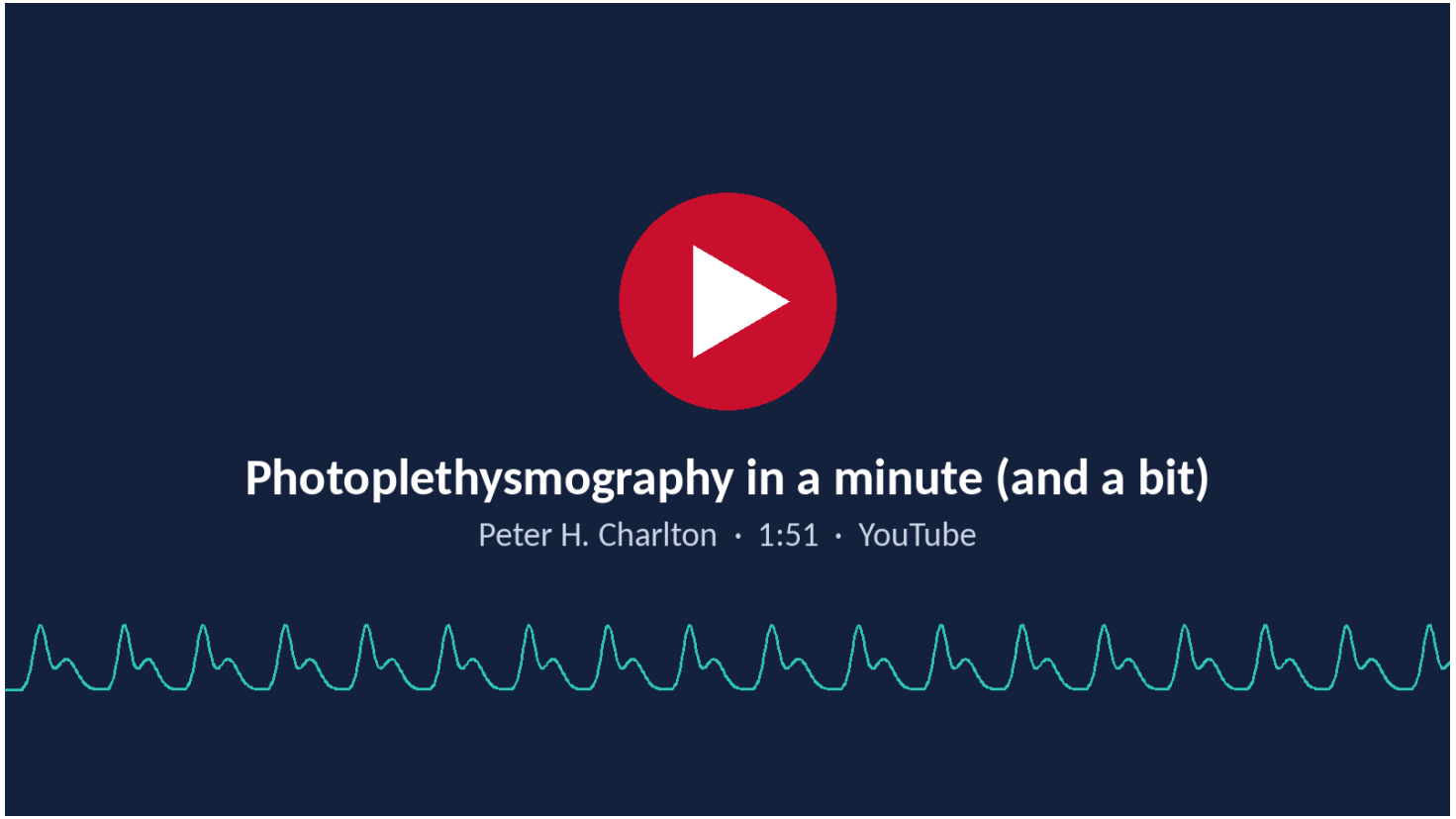
Sensor construction also limits your choice: many biofeedback PPG sensors fit only one site (for example an ear clip or a finger housing).

Elgendi, M. (2012). On the analysis of fingertip photoplethysmogram signals. Curr Cardiol Rev, 8(1), 14–25. doi:10.2174/157340312801215782

Allen, J. (2007). Photoplethysmography and its application in clinical physiological measurement. Physiol Meas, 28(3), R1–R39. doi:10.1088/0967-3334/28/3/R01

Přibil, J., et al. (2020). Comparative measurement of the PPG signal on different human body positions by sensors working in reflection and transmission modes. Eng Proc, 2(1), 69. doi:10.3390/ecsa-7-08204

Video: photoplethysmography explained in under two minutes



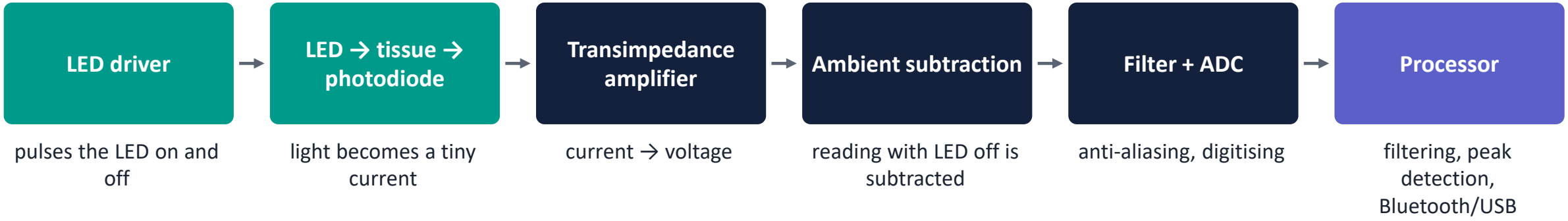
If the video does not play here, open it on YouTube

While you watch, look for:

- What the pulsatile and the steady parts of the PPG signal are.
- Where PPG sensors are used today, from hospitals to smartwatches.
- Which body information, besides heart rate, a PPG signal can carry.

Video: Charlton, P. H. Photoplethysmography in a minute (and a bit). YouTube. https://www.youtube.com/watch?v=HnXDvN4WNX8

Inside a PPG sensor: a small optical measurement system



- LEDs are **pulsed**: the sensor samples once with the LED on and once with it off, and subtracts the two. This cancels room light and saves battery. It is why smartwatch LEDs flicker.
- Wearables add an **accelerometer** to detect and remove motion (slide 31).
- Integrated **analog front-end chips** combine LED drivers, amplifiers, ambient cancellation and ADC in one package.
- A clinical finger clip and a smartwatch share this architecture; they differ in optics, site and processing.

Tamura, T., et al. (2014). Wearable photoplethysmographic sensors: Past and present. Electronics, 3(2), 282–302. doi:10.3390/electronics3020282

Charlton, P. H., et al. (2022). Wearable photoplethysmography for cardiovascular monitoring. Proc IEEE, 110(3), 355–381. doi:10.1109/JPROC.2022.3149785

Optional at home: Mould, S. The bizarre flashing lights on a smartwatch [video]. YouTube. https://www.youtube.com/watch?v=BFZxlauizx0

Fingers, thumb and earlobe are the classic PPG sites; each has trade-offs

Site	Signal	Motion robustness	Effect of cold or stress	Notes
Finger (pad side, one segment)	High	Low	High	Standard biofeedback site; left and right hands are equivalent
Thumb	Highest	Low	Moderate	Use when fingers are small or poorly perfused
Earlobe	Lower (fewer, smaller vessels)	Unaffected by hand movement; the clip itself can shift	Lower	Closer to the heart; less finger vascular-tone rhythm in spectral HRV
Wrist (reflectance)	Lower	Moderate, with accelerometer	Moderate	Smartwatches: fine at rest, error grows with activity



Transmission finger clip (pulse oximeter). Photo: “Project 365 #154: 030614 Saturation” by comedy_nose, Flickr, public domain

Přibil, J., et al. (2020). Comparative measurement of the PPG signal on different human body positions by sensors working in reflection and transmission modes. Eng Proc, 2(1), 69. doi:10.3390/ecsa-7-08204

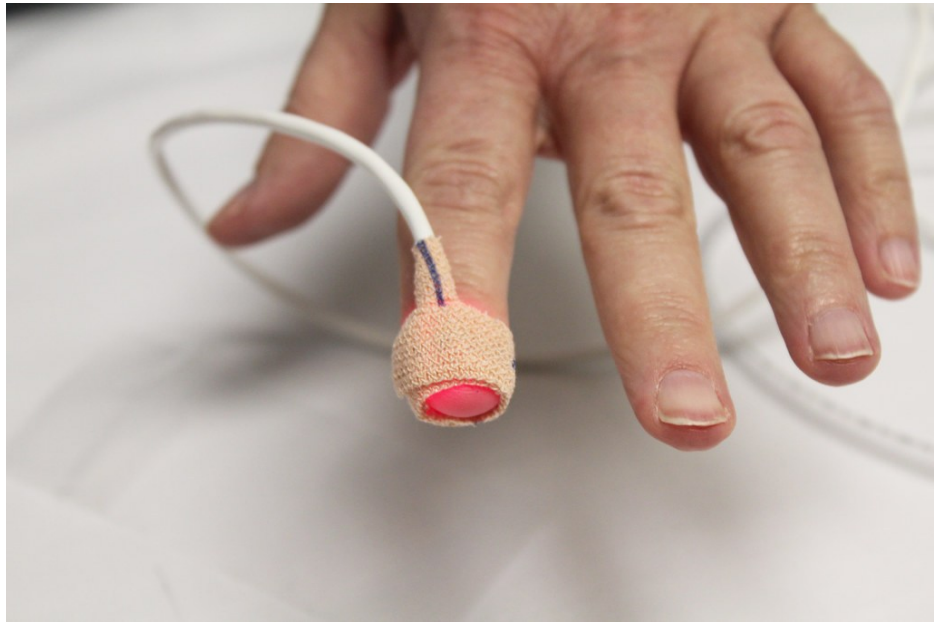
Peper, E., et al. (2010). Garbage in; garbage out: Identify blood volume pulse (BVP) artifacts before analyzing and interpreting BVP, BVP amplitude, and HR/RSA data. Biofeedback, 38(1), 19–23. doi:10.5298/1081-5937-38.1.19

Bent, B., et al. (2020). Investigating sources of inaccuracy in wearable optical heart rate sensors. npj Digit Med, 3(1), 18. doi:10.1038/s41746-020-0226-6

Shaffer, F., Khazan, I., & Moss, D. (2024). HRV instrumentation. BCIA HRV Biofeedback Didactic Workshop [lecture slides], practice recommendations.

Small placement details decide whether a PPG recording is usable

- Attach to the **pad (palm) side** of a larger finger or the thumb, within **one finger segment**.
- **Snug, not tight:** an elastic band or self-adhesive wrap. Too tight squeezes the vessels (pressure artifact).
- Hand **relaxed, still and at heart level**, e.g. resting on the thigh or an armrest.
- **Shield from room light** with a dark cloth, sock or wrap.
- Remove **dark nail polish** (blocks light); **warm cold hands** first.
- **No skin preparation** beyond clean, dry hands: PPG is optical, so electrical contact does not matter.

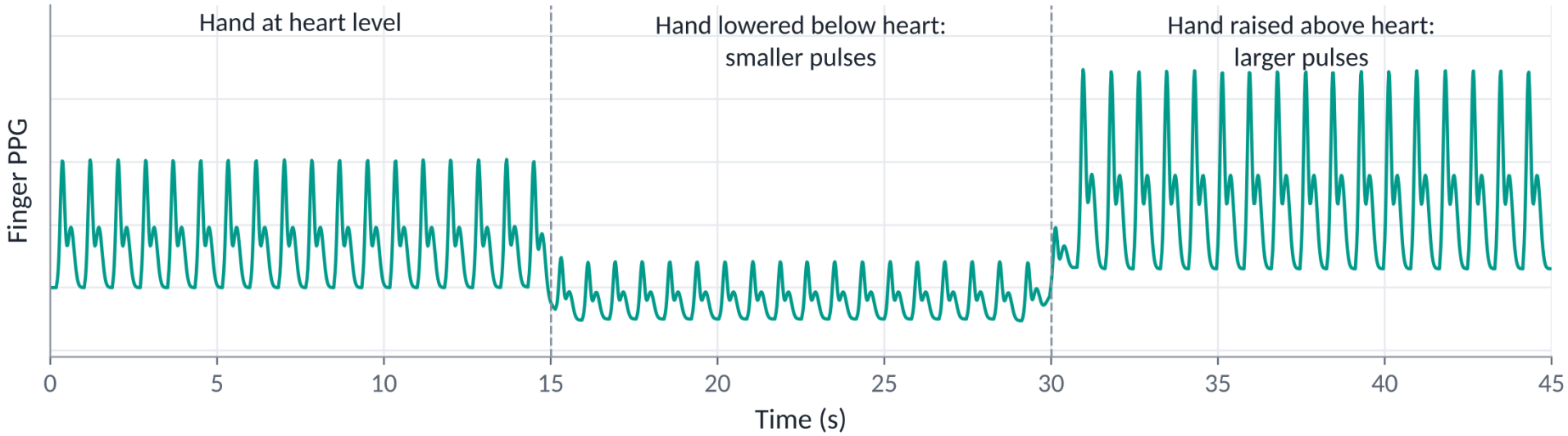


Adhesive PPG sensor on a fingertip. Photo: “Pulse oximeter” by quinn.anya, Flickr, CC BY-SA 2.0

Peper, E., et al. (2010). Garbage in; garbage out: Identify blood volume pulse (BVP) artifacts before analyzing and interpreting BVP, BVP amplitude, and HR/RSA data. Biofeedback, 38(1), 19–23. doi:10.5298/1081-5937-38.1.19

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Moving the hand up or down changes pulse amplitude and baseline

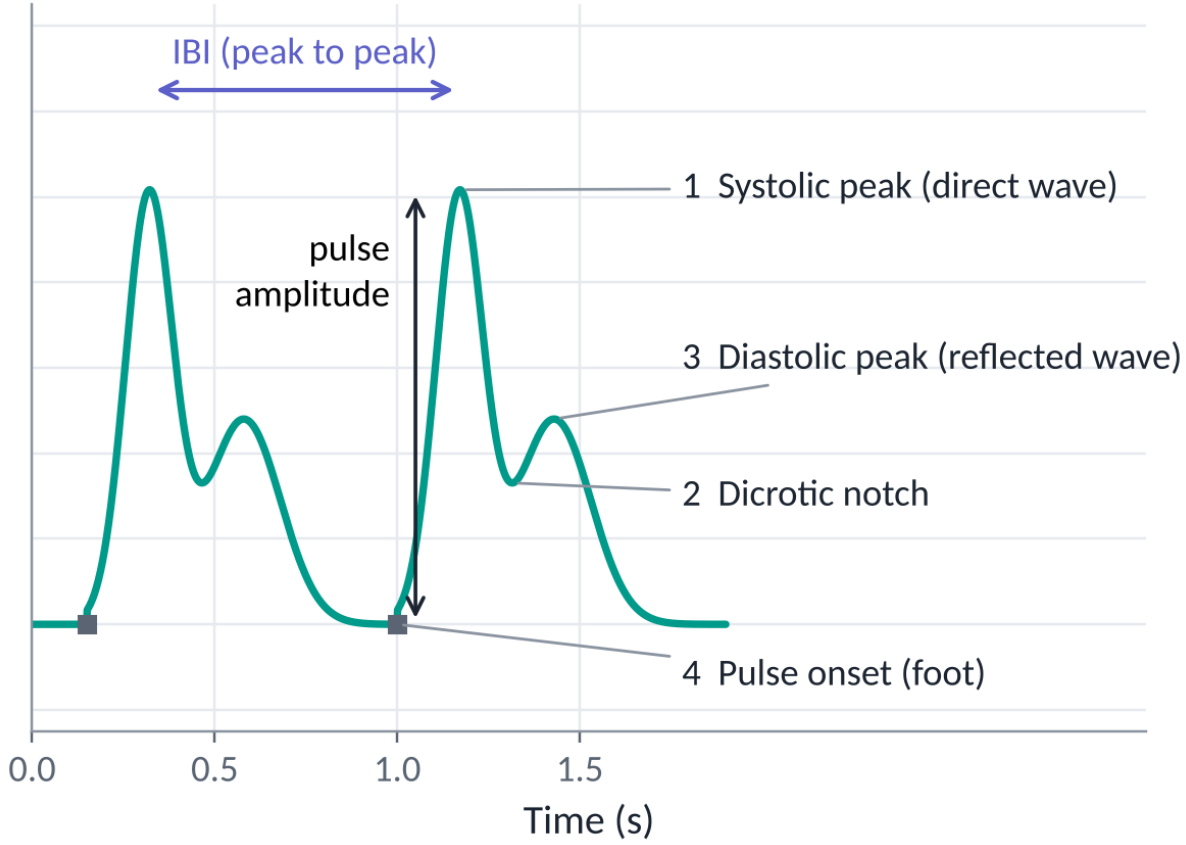


- Hand height changes **hydrostatic pressure** in the hand, altering venous filling and arterial wall tension.
- In a controlled study, finger PPG amplitude **fell when the hand was lowered** below heart level and **rose when it was raised** above it (Hickey et al., 2015). Some practitioner guides describe the opposite direction; either way, recordings at different heights are **not comparable**.
- **Practical rule:** keep the sensor at heart level and the arm still for the whole session.

Hickey, M., et al. (2015). The effect of vascular changes on the photoplethysmographic signal at different hand elevations. Physiol Meas, 36(3), 425–440. doi:10.1088/0967-3334/36/3/425

Peper, E., et al. (2010). Garbage in; garbage out: Identify blood volume pulse (BVP) artifacts before analyzing and interpreting BVP, BVP amplitude, and HR/RSA data. Biofeedback, 38(1), 19–23. doi:10.5298/1081-5937-38.1.19

A PPG pulse has a characteristic shape; its peak or foot marks each beat



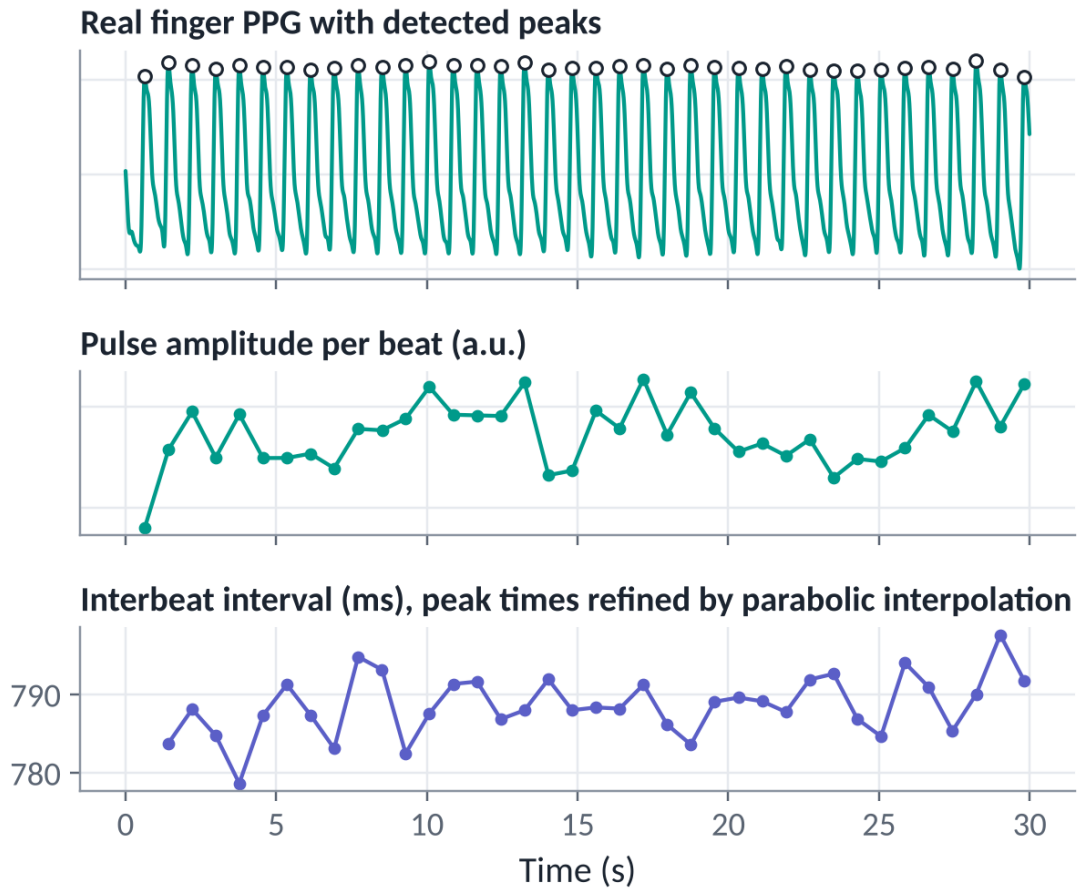
- **1 Systolic peak:** volume maximum from the direct pressure wave after the ventricles eject.
- **2 Dicrotic notch:** small dip between the direct and the reflected wave (linked to aortic valve closure and wave reflection).
- **3 Diastolic peak:** wave reflected back from the lower body and vessel branchings.
- **4 Foot (onset):** start of the upstroke; often the most stable timing point.
- **Pulse amplitude** (peak minus foot) tracks local blood flow and shrinks with vasoconstriction.
- The notch fades with **age and arterial stiffness**; it sits lower with vasodilation and higher with vasoconstriction.

Elgendi, M. (2012). On the analysis of fingertip photoplethysmogram signals. Curr Cardiol Rev, 8(1), 14–25. doi:10.2174/157340312801215782

Allen, J. (2007). Photoplethysmography and its application in clinical physiological measurement. Physiol Meas, 28(3), R1–R39. doi:10.1088/0967-3334/28/3/R01

From one PPG signal we get intervals, amplitudes and their variability

- **Interbeat interval (IBI):** time between successive peaks or feet. Gives pulse rate and **pulse rate variability (PRV)**.
- **Pulse amplitude:** peak-to-trough height per beat, in arbitrary units, comparable only within one recording.
- **Pulse amplitude variability (PAV):** beat-to-beat changes in amplitude, reflecting changes in vascular tone (sympathetic activity, breathing, temperature).
- **Sub-sample timing:** at 125 Hz raw peak times move in 8 ms steps. Fitting a parabola through the three samples around each peak refines the time (used in this figure).

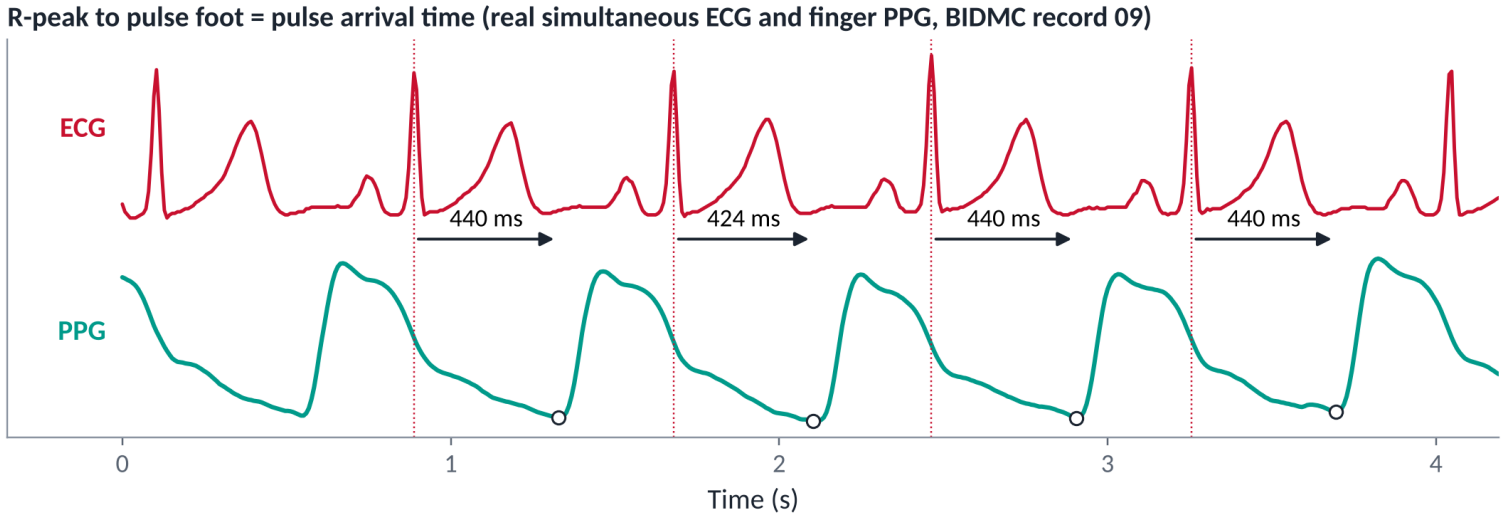


Allen, J. (2007). Photoplethysmography and its application in clinical physiological measurement. Physiol Meas, 28(3), R1–R39. doi:10.1088/0967-3334/28/3/R01

Data: BIDMC PPG and Respiration Dataset, PhysioNet (Pimentel et al., 2017; Goldberger et al., 2000).

Pimentel, M. A. F., et al. (2017). Toward a robust estimation of respiratory rate from pulse oximeters. IEEE Trans Biomed Eng, 64(8), 1914–1923. doi:10.1109/TBME.2016.2613124

Pulse rate variability is a close cousin of HRV, not an exact copy



- The pulse reaches the finger a few hundred ms after the R-peak: **pulse arrival time (PAT)** = pre-ejection period + pulse transit time.
- PAT changes from beat to beat with blood pressure, vessel stiffness, breathing and posture, so **PPG intervals do not exactly equal ECG intervals**.
- In healthy people at rest, PRV agrees well with HRV for mean intervals and slower components; agreement **worsens for short-term measures (RMSSD, HF)**, during movement or stress, and with poor signals.
- Use **ECG** when beat-exact timing matters (research, arrhythmias); PPG is fine for training at rest.

Schäfer, A., & Vagedes, J. (2013). How accurate is pulse rate variability as an estimate of heart rate variability? Int J Cardiol, 166(1), 15–29. doi:10.1016/j.ijcard.2012.03.119

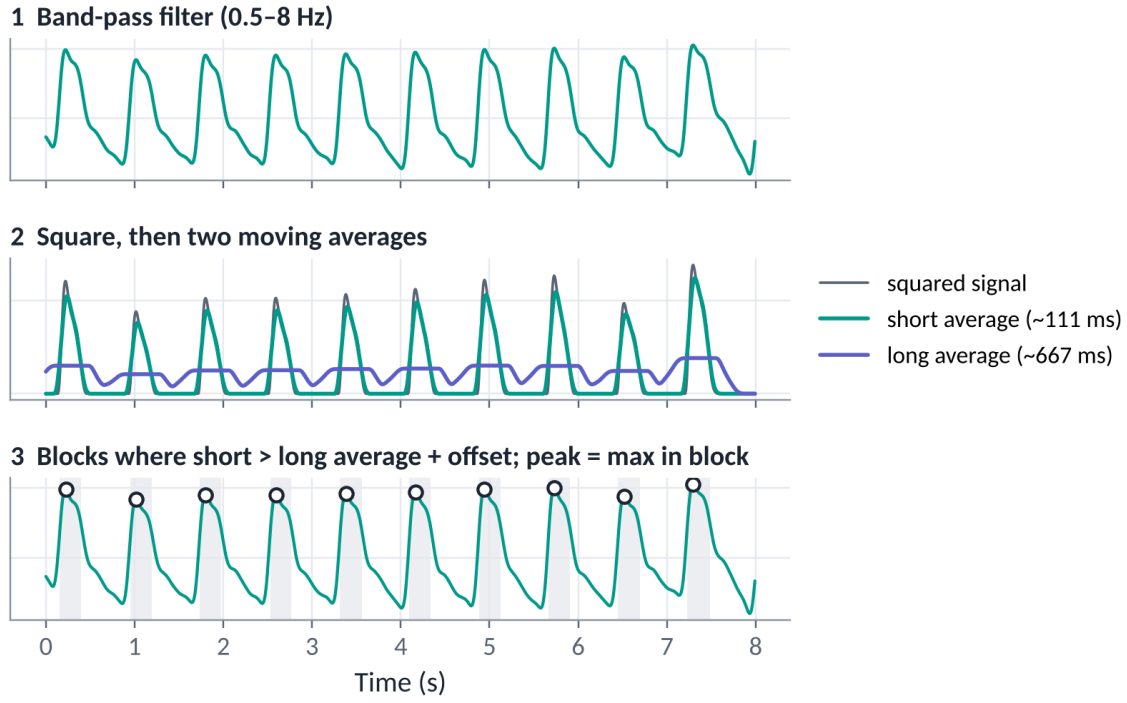
Mejía-Mejía, E., et al. (2020). Pulse rate variability in cardiovascular health: A review on its applications and relationship with heart rate variability. Physiol Meas, 41(7), 07TR01. doi:10.1088/1361-6579/ab998c

Mukkamala, R., et al. (2015). Toward ubiquitous blood pressure monitoring via pulse transit time: Theory and practice. IEEE Trans Biomed Eng, 62(8), 1879–1901. doi:10.1109/TBME.2015.2441951

Data: BIDMC PPG and Respiration Dataset, PhysioNet (Pimentel et al., 2017; Goldberger et al., 2000).

Engineering view: sample fast enough, filter, then detect each pulse

- **Sampling rate:** in a pilot study at rest, PRV measures did not differ significantly down to about 25 Hz; around 100 Hz or more gives margin for noisy data, and interpolation refines timing.
- **Filtering:** band-pass about 0.5–8 Hz removes baseline drift and high-frequency noise.
- **Elgendi detector:** square the positive part of the signal; compare a short moving average (≈ 111 ms, one systolic peak) with a long one (≈ 667 ms, one beat) plus a small offset (0.02 × mean of the squared signal). Blocks where short exceeds this threshold, and that are at least 111 ms long, each contain one peak: its maximum.




```
import neurokit2 as nk

clean = nk.ppg_clean(ppg, sampling_rate=fs)
info = nk.ppg_findpeaks(clean, sampling_rate=fs,
                        method="elgendi")
peaks = info["PPG_Peaks"]
```

Choi, A., & Shin, H. (2017). Photoplethysmography sampling frequency: Pilot assessment of how low can we go to analyze pulse rate variability with reliability? Physiol Meas, 38(3), 586–600. doi:10.1088/1361-6579/aa5efa

Elgendi, M., et al. (2013). Systolic peak detection in acceleration photoplethysmograms measured from emergency responders in tropical conditions. PLoS ONE, 8(10), e76585. doi:10.1371/journal.pone.0076585

Makowski, D., et al. (2021). NeuroKit2: A Python toolbox for neurophysiological signal processing. Behav Res Methods, 53(4), 1689–1696. doi:10.3758/s13428-020-01516-y

An artifact is anything in the signal that is not the heartbeat you want to measure

Body

Ectopic beats, atrial fibrillation, cold- or stress-induced vasoconstriction.

Behaviour

Moving, talking, tensing muscles, changing posture.

Environment

Room light, mains interference (50 Hz in Europe), phones and monitors.

Hardware and set-up

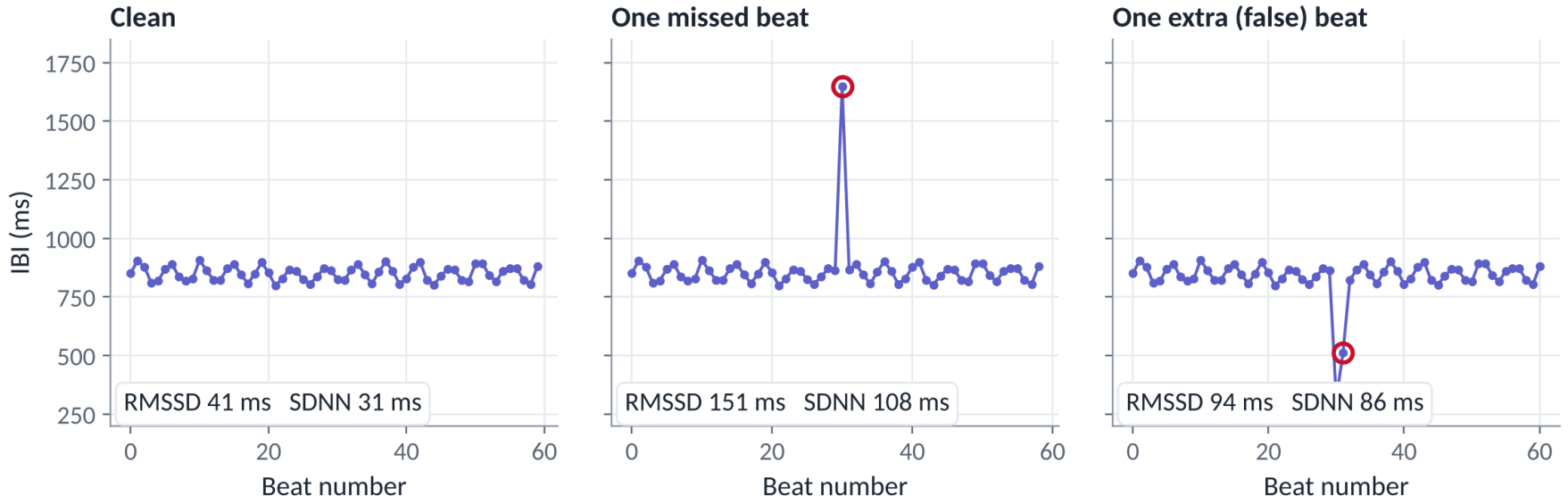
Loose or over-tight sensor, light leaks, poor electrode contact, saturated amplifier.

Prevention beats correction. Fix the set-up before recording, then inspect the **raw waveform**, not just the heart-rate number. Artifacts that are obvious in the raw signal can be invisible in a summary value.

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Peper, E., et al. (2010). Garbage in; garbage out: Identify blood volume pulse (BVP) artifacts before analyzing and interpreting BVP, BVP amplitude, and HR/RSA data. Biofeedback, 38(1), 19–23. doi:10.5298/1081-5937-38.1.19

One missed or one extra beat can more than double RMSSD



- **Missed beat:** two intervals merge into one long interval; HR briefly drops. **Extra (false) beat:** one interval splits into two short ones; HR briefly jumps.
- Both create large successive differences, so short-term HRV (RMSSD, HF power) is **inflated, not reduced**. Here RMSSD rises from 41 ms to 151 ms (missed) and 94 ms (extra).
- Common practice: inspect every segment; if more than about **5% of intervals** are artifactual, discard the segment rather than correct it.

Berntson, G. G., & Stowell, J. R. (1998). ECG artifacts and heart period variability: Don't miss a beat! Psychophysiology, 35(1), 127–132. doi:10.1111/1469-8986.3510127

Berntson, G. G., et al. (1997). Heart rate variability: Origins, methods, and interpretive caveats. Psychophysiology, 34(6), 623–648. doi:10.1111/j.1469-8986.1997.tb02140.x

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Three habits catch most artifacts before they reach your analysis

1 Know your baseline

Record yourself regularly so you know what normal signals and values look like on your own equipment.

2 Run a tracking test

Ask the participant to breathe slowly in and out. Instantaneous HR should rise on inhalation and fall on exhalation. If not, check the sensor.

3 Inspect the raw waveform

Look at the raw signal, not only derived HR or HRV. Many artifacts are obvious in the waveform.

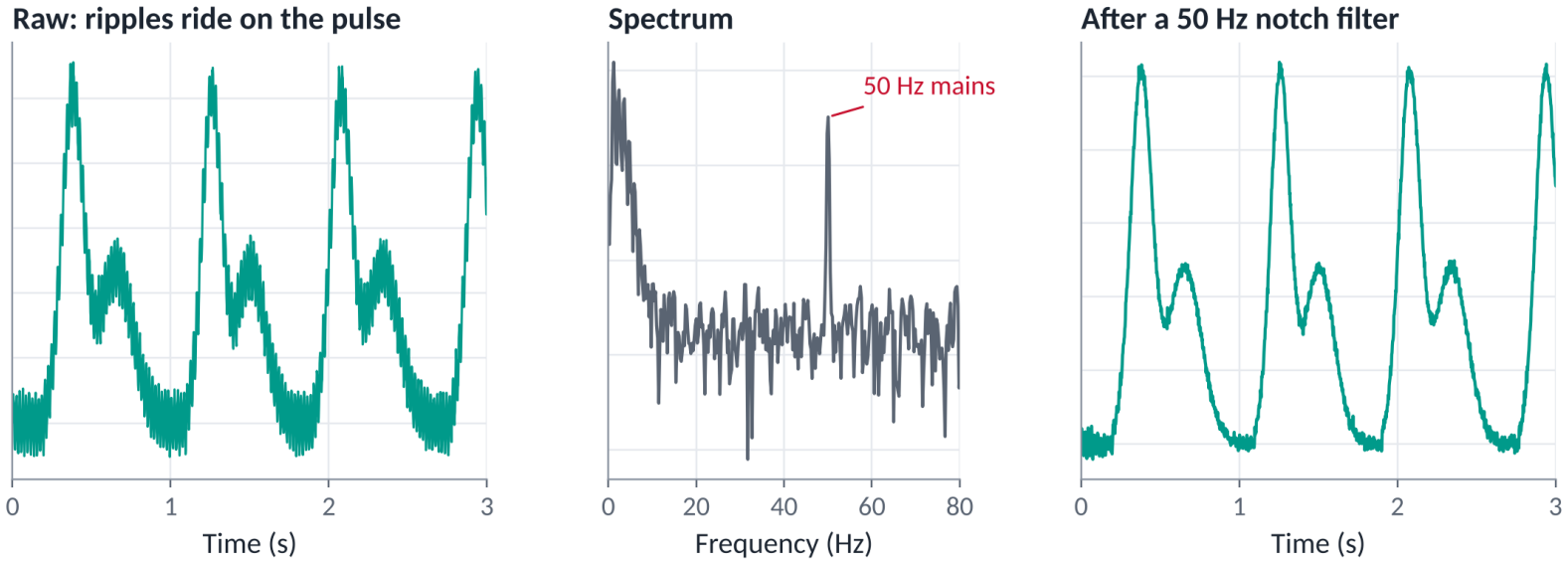
Learn artifact shapes by creating them on purpose: move the finger, tap, uncover the sensor, squeeze the band, tense the arm, and watch what happens to the raw signal and to the HRV values. It does not harm the equipment.

Peper, E., et al. (2010). Garbage in; garbage out: Identify blood volume pulse (BVP) artifacts before analyzing and interpreting BVP, BVP amplitude, and HR/RSA data. Biofeedback, 38(1), 19–23. doi:10.5298/1081-5937-38.1.19

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Shaffer, F., Khazan, I., & Moss, D. (2024). HRV instrumentation. BCIA HRV Biofeedback Didactic Workshop [lecture slides], practice recommendations.

Mains interference (50 Hz in Europe) shows up as fine ripples on the pulse

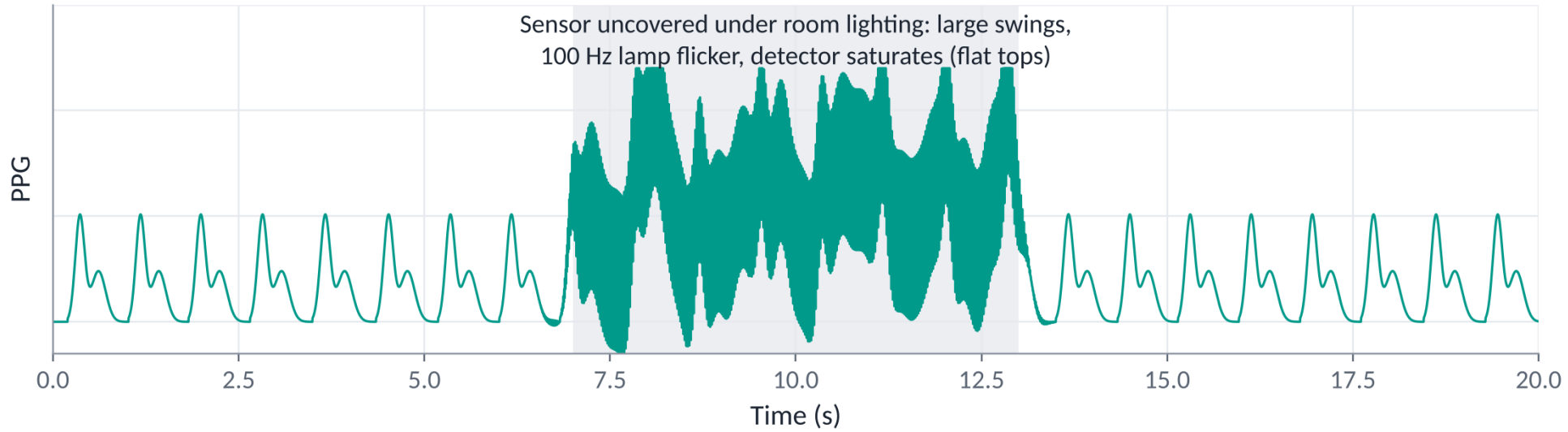


- Power lines, chargers and lamps couple **50 Hz** (60 Hz in the Americas) into cables and electronics. PPG is less sensitive than ECG because it is optical, but its electronics can still pick it up.
- **Fixes:** enable the 50 Hz notch filter; keep the encoder or amplifier about **1 m** from other electronics; unplug unused sensor cables; inspect the raw signal.
- **Engineering note:** a notch filter removes a narrow band (here Q = 30, about 1.7 Hz wide at 50 Hz) and leaves the pulse shape intact.

Elgendi, M. (2012). On the analysis of fingertip photoplethysmogram signals. Curr Cardiol Rev, 8(1), 14–25. doi:10.2174/157340312801215782

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Room light reaching the photodiode can swamp the pulse

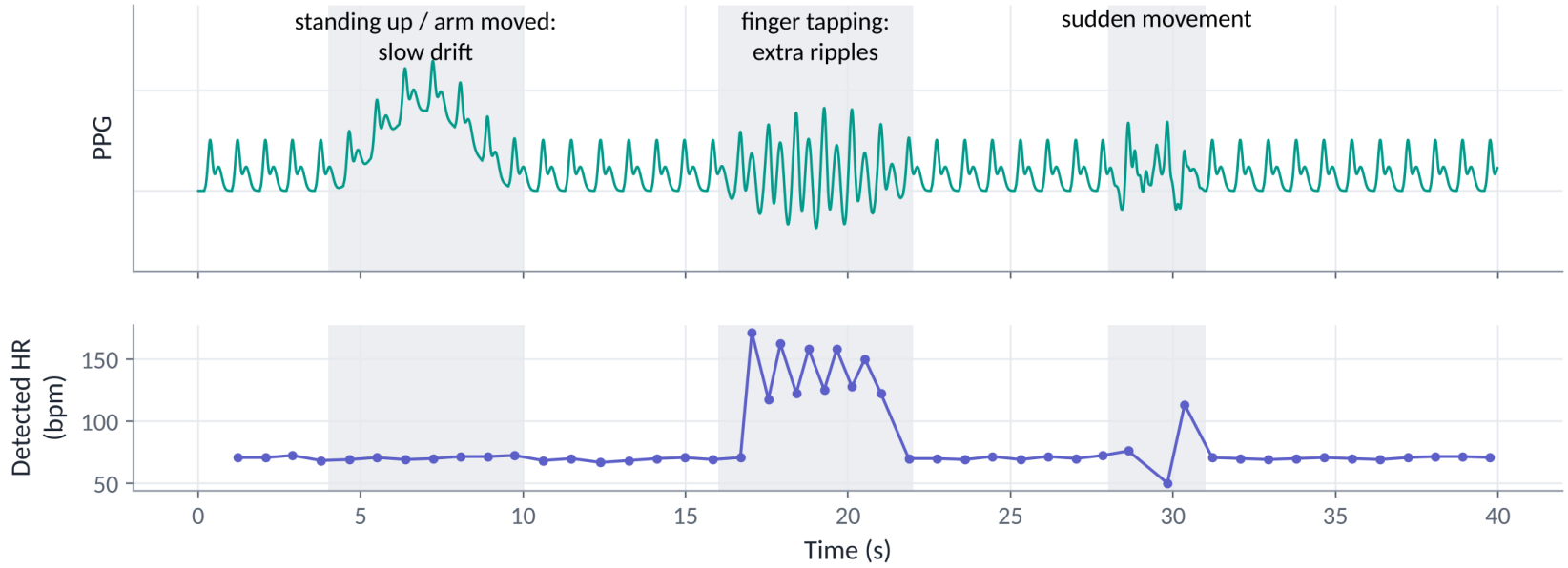


- Daylight, lamps or screens leaking under the sensor add large, irregular changes and can **saturate** the detector (flat tops).
- Many lamps flicker at **twice the mains frequency (100 Hz in Europe)**, adding a fast ripple.
- **Fixes:** cover the sensor (dark cloth, sock or self-adhesive wrap); avoid direct light on the hand; keep the hand still so gaps do not open; check the raw signal. In hardware, LED on/off ambient subtraction (slide 17) helps.

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Tamura, T., et al. (2014). Wearable photoplethysmographic sensors: Past and present. Electronics, 3(2), 282–302. doi:10.3390/electronics3020282

Movement is the main cause of distorted PPG signals



- Sensor movement changes the optical path and lets room light in; posture changes shift blood volume.
- Shapes vary: **slow drifts** (standing up, arm moved), **extra ripples** that look like additional notches (finger tapping), **sudden large swings**.
- Consequence: the detector counts **false beats or misses real ones**, and heart rate jumps (lower panel).

Elgendi, M. (2012). On the analysis of fingertip photoplethysmogram signals. Curr Cardiol Rev, 8(1), 14–25. doi:10.2174/157340312801215782

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

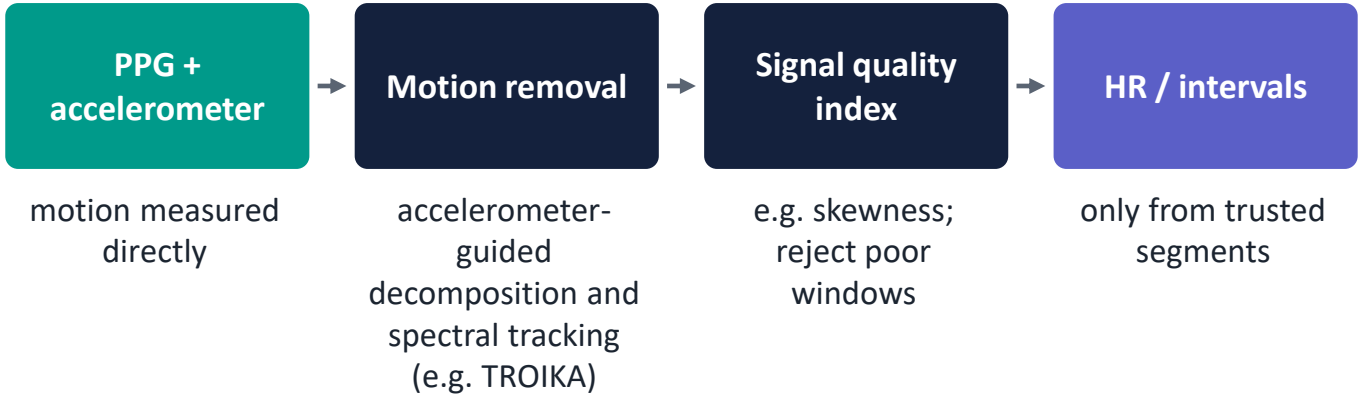
Peper, E., et al. (2010). Garbage in; garbage out: Identify blood volume pulse (BVP) artifacts before analyzing and interpreting BVP, BVP amplitude, and HR/RSA data. Biofeedback, 38(1), 19–23. doi:10.5298/1081-5937-38.1.19

Prevent movement artifacts at set-up; wearables also remove them in software

In the lab

- Attach the sensor firmly (or use a rigid housing); rest the hand on the thigh or an armrest.
- Tape the cable to clothing for **strain relief**.
- Ask the participant to keep still, and **check** that they do.
- Inspect the raw signal throughout.

In wearables: algorithms



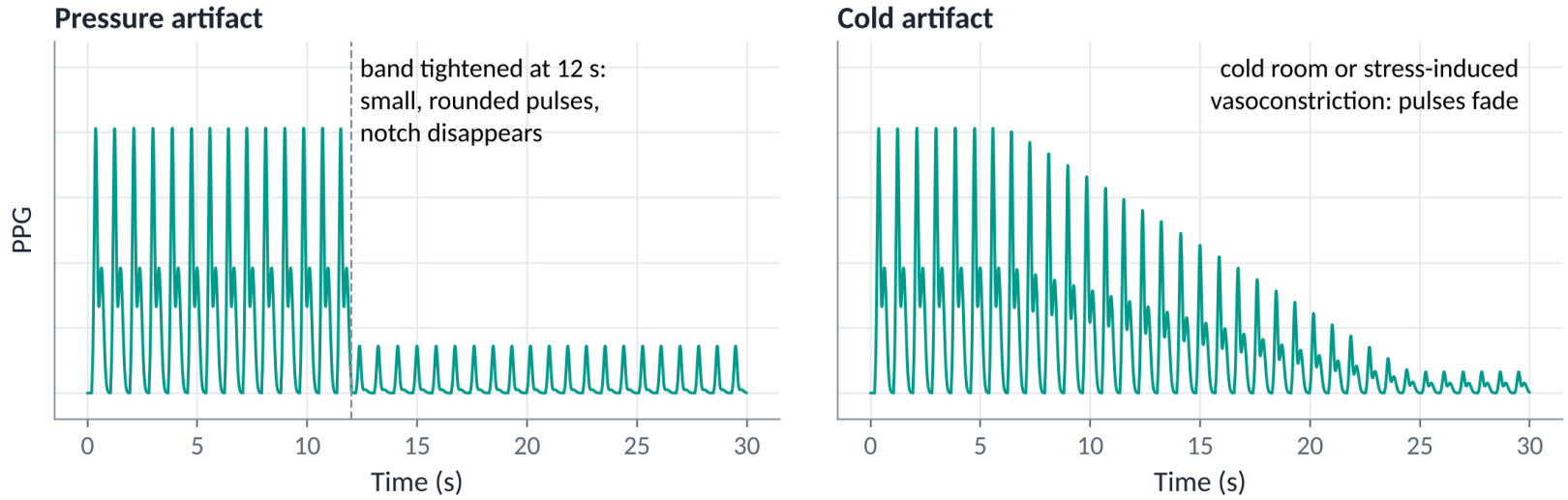
Even good wrist sensors lose accuracy during activity: in one study, heart-rate error was on average about 30% higher during activity than at rest.

Zhang, Z., et al. (2015). TROIKA: A general framework for heart rate monitoring using wrist-type photoplethysmographic signals during intensive physical exercise. IEEE Trans Biomed Eng, 62(2), 522–531. doi:10.1109/TBME.2014.2359372

Elgendi, M. (2016). Optimal signal quality index for photoplethysmogram signals. Bioengineering, 3(4), 21. doi:10.3390/bioengineering3040021

Bent, B., et al. (2020). Investigating sources of inaccuracy in wearable optical heart rate sensors. npj Digit Med, 3(1), 18. doi:10.1038/s41746-020-0226-6

Too much pressure or cold hands shrink the pulse until beats are lost



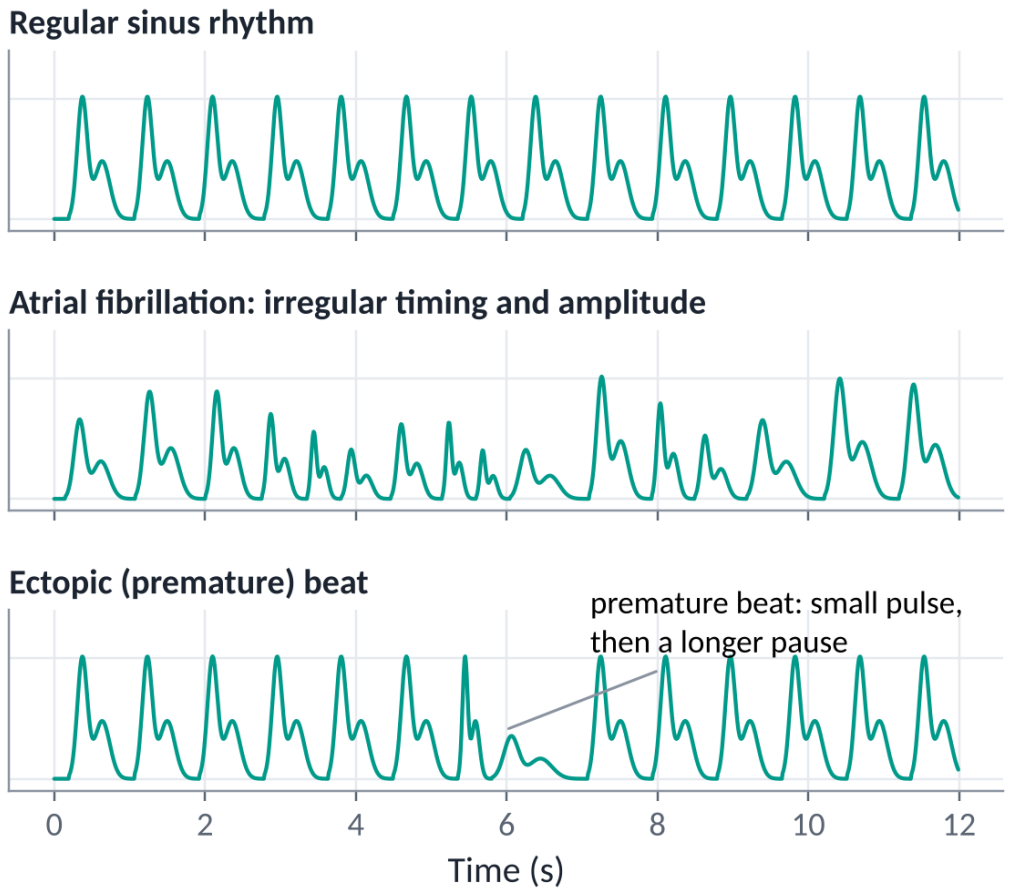
Artifact	Cause	What you see	Fix
Pressure	Band too tight; hand resting on the sensor	Small, rounded pulses; notch disappears; may go flat	Loosen the band; move weight off the sensor
Cold	Cool room, cold hands, stress-induced vasoconstriction	Pulses gradually fade	Room at least 23 °C; warm the hands (warm water, a few minutes of relaxation); thumb or earlobe; or use ECG

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Peper, E., et al. (2010). Garbage in; garbage out: Identify blood volume pulse (BVP) artifacts before analyzing and interpreting BVP, BVP amplitude, and HR/RSA data. Biofeedback, 38(1), 19–23. doi:10.5298/1081-5937-38.1.19

Irregular rhythms are real physiology, not sensor faults, but they still break HRV analysis

- **Atrial fibrillation (AF):** disorganised atrial activity; pulses arrive at irregular intervals, with varying and often smaller amplitude and a faster rate.
- **Ectopic beat:** an early (premature) beat with a small pulse, followed by a longer pause.
- Standard HRV assumes **sinus rhythm**, so these intervals must be identified and excluded, or the segment discarded.
- **Opportunity:** smartwatches use exactly these PPG irregularities to screen for AF; the Apple Heart Study enrolled 419,297 participants.



Pereira, T., et al. (2020). Photoplethysmography based atrial fibrillation detection: A review. npj Digit Med, 3(1), 3. doi:10.1038/s41746-019-0207-9

Perez, M. V., et al. (2019). Large-scale assessment of a smartwatch to identify atrial fibrillation. N Engl J Med, 381(20), 1909–1917. doi:10.1056/NEJMoa1901183

Optical sensing is not equally accurate for everyone: an engineering responsibility

Physics

Melanin absorbs visible light, so darker skin gives a weaker pulsatile signal. Fallow et al. (2013) found lower modulation in darker skin at all wavelengths tested; green still worked best.

Evidence of bias

Pulse oximeters missed hidden low oxygen about 3 times more often in Black than in White patients (Sjoding et al., 2020).

Heart rate

A study of wearables found error depended on device and activity, with no significant difference across skin tones (Bent et al., 2020). Test it; do not assume.

- Tattoos, nail polish and low perfusion also reduce optical signal quality.
- **Design takeaway:** validate devices and algorithms on diverse participants, and report performance by subgroup.

Fallow, B. A., et al. (2013). Influence of skin type and wavelength on light wave reflectance. J Clin Monit Comput, 27(3), 313–317. doi:10.1007/s10877-013-9436-7

Sjoding, M. W., et al. (2020). Racial bias in pulse oximetry measurement. N Engl J Med, 383(25), 2477–2478. doi:10.1056/NEJMc2029240

Bent, B., et al. (2020). Investigating sources of inaccuracy in wearable optical heart rate sensors. npj Digit Med, 3(1), 18. doi:10.1038/s41746-020-0226-6

When the PPG signal is weak or noisy, work through these checks in order

1 **Hand moving or sensor loose?**

Yes: Re-attach firmly, support the arm, ask for stillness

2 **Room light reaching the sensor?**

Yes: Cover the sensor; avoid direct light

3 **Band too tight?**

Yes: Loosen it; take weight off the sensor

4 **Pulses fading, hands cold?**

Yes: Warm the hands; allow a few minutes to relax

5 **Still weak on a finger?**

Yes: Move to the thumb or the earlobe

6 **Still unusable?**

Yes: Switch to ECG

If the answer is no, go to the next check.

Peper, E., et al. (2010). Garbage in; garbage out: Identify blood volume pulse (BVP) artifacts before analyzing and interpreting BVP, BVP amplitude, and HR/RSA data. Biofeedback, 38(1), 19–23. doi:10.5298/1081-5937-38.1.19

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

PPG artifacts at a glance

Artifact	Looks like	Main cause	Prevention or fix
Mains (50 Hz)	Fine ripples; 50 Hz spectral peak	Nearby electronics and cables	Notch filter; 1 m distance; unplug unused cables
Room light	Large irregular swings, flat tops, 100 Hz ripple	Uncovered or loose sensor	Cover the sensor; avoid direct light
Movement	Drifts, extra ripples, sudden swings; false or missed beats	Hand or finger moving, cable pulling	Firm attachment, strain relief, instructions; accelerometer in wearables
Pressure	Small, rounded pulses; notch lost	Band too tight, weight on sensor	Loosen the band
Cold	Fading pulses	Cool room, vasoconstriction	Warm room and hands; thumb or earlobe; ECG
Irregular rhythm	Irregular intervals and amplitudes	AF, ectopic beats	Identify; exclude intervals or discard the segment

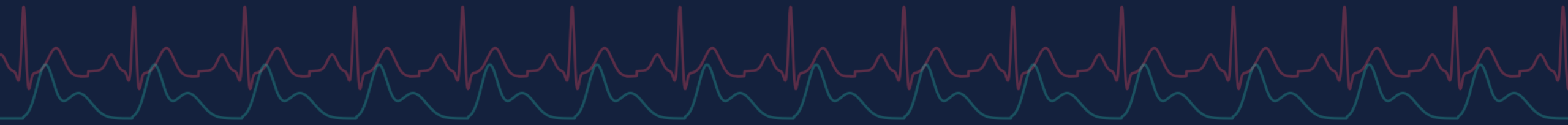
Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Elgendi, M. (2012). On the analysis of fingertip photoplethysmogram signals. Curr Cardiol Rev, 8(1), 14–25. doi:10.2174/157340312801215782

PART 2

The electrocardiogram (ECG)

The reference method for beat timing: electrodes, amplifiers, placements, beat detection and artifacts



ECG gives the most accurate beat timing; PPG is quicker to set up

	ECG	PPG
Timing reference	Sharp R-peak (QRS lasts about 0.1 s or less)	Rounded pulse, delayed by pulse travel time
Accuracy for HRV	Reference method; recommended for research	Acceptable for training and monitoring at rest
Set-up	Skin preparation, disposable electrodes, sometimes clothing moved	Clip or wrap, a few seconds
Sensitive to	Mains, muscle (EMG), electrode contact, polarity	Movement, room light, cold, pressure
Not affected by	Cold hands, room light	Skin-electrode impedance

Rule of thumb: publishing research or analysing arrhythmias? Use ECG. Training or monitoring at rest? A clean PPG recording is usually enough.

Quigley, K. S., et al. (2024). Publication guidelines for human heart rate and heart rate variability studies in psychophysiology, Part 1. Psychophysiology, 61(9), e14604. doi:10.1111/psyp.14604

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Schäfer, A., & Vagedes, J. (2013). How accurate is pulse rate variability as an estimate of heart rate variability? Int J Cardiol, 166(1), 15–29. doi:10.1016/j.ijcard.2012.03.119

An ECG electrode is an electrochemical interface: low, balanced impedance keeps the signal clean

- Standard biopotential electrodes are **silver/silver chloride (Ag/AgCl)** with conductive gel, snapped onto lead wires.
- The interface behaves like a small battery (**half-cell potential**) plus a resistor and capacitor in parallel, in series with gel and skin resistance (circuit, right).
- The dry outer skin layer has high impedance. **Cleaning with an alcohol wipe** (and gentle abrasion if needed) lowers it.
- **Pre-gelled disposable** electrodes: fast, hygienic, low impedance. **Dry** electrodes (chest straps) work once sweat builds up.
- Aim for impedance **below about 10 kΩ**, with electrodes **within about 5 kΩ** of each other.

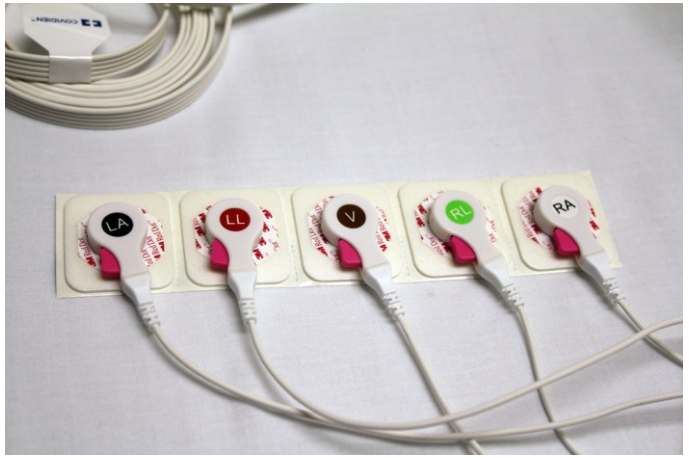
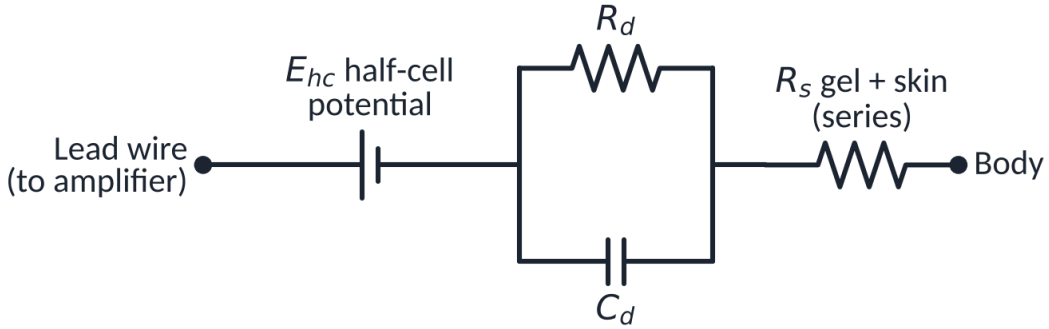


Photo: “Monitoring electrode” by quinn.anya, Flickr, CC BY-SA 2.0



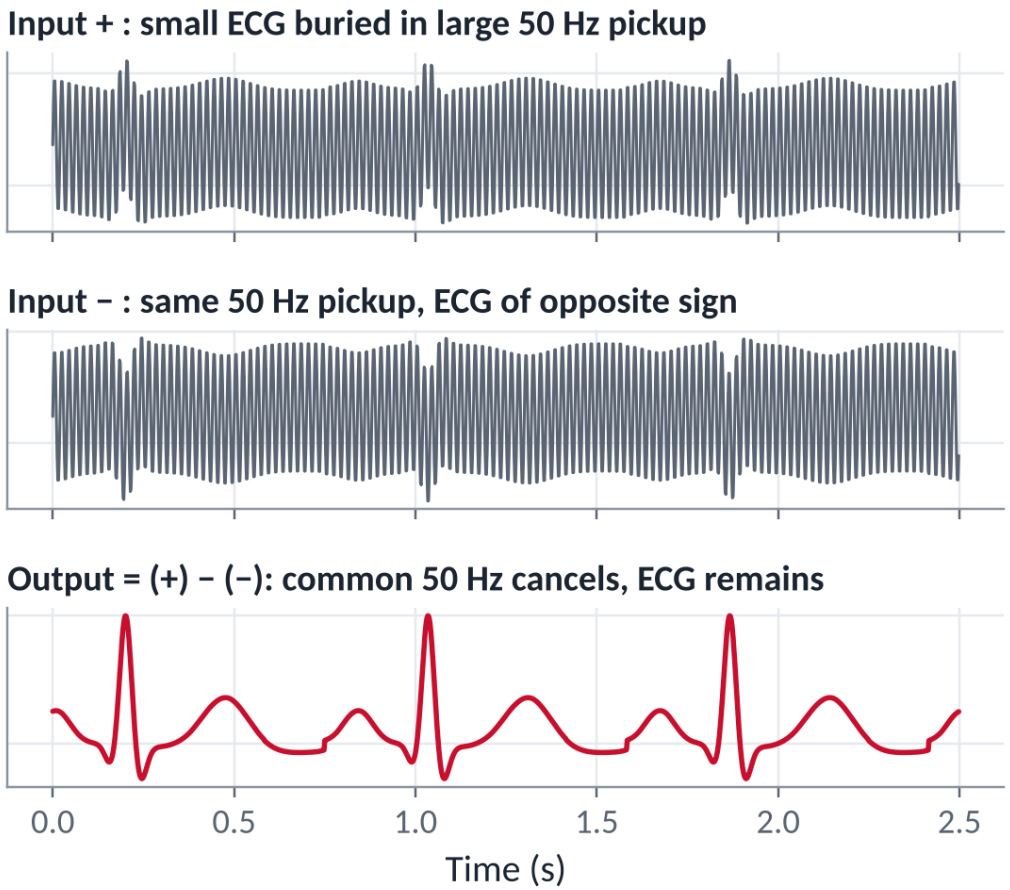
Webster, J. G. (Ed.) (2010). Medical instrumentation: Application and design (4th ed.). Wiley.

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Shaffer, F., Khazan, I., & Moss, D. (2024). HRV instrumentation. BCIA HRV Biofeedback Didactic Workshop [lecture slides], practice recommendations.

Two active electrodes and a reference let a differential amplifier cancel common noise

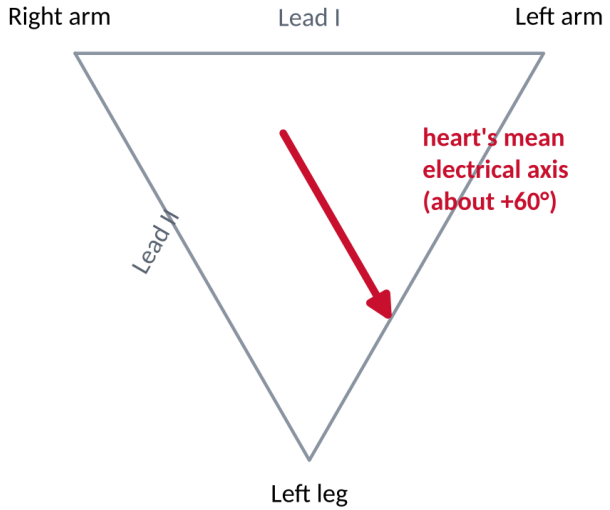
- The amplifier measures the **difference** between the two active electrodes (+ and −). The ECG differs between them; mains interference picked up by the body is almost **identical at both**.
- Subtraction removes this **common-mode** signal. The **common-mode rejection ratio (CMRR)**, often 100 dB or more in good ECG amplifiers, says how well.
- The third electrode (reference/ground) ties the body to the amplifier reference; a **driven-right-leg** circuit goes further and actively lowers the common-mode voltage.
- **Unequal electrode impedances** convert common-mode noise into a differential error: one more reason for good skin preparation.
- Three electrodes are enough for HRV; a diagnostic 12-lead ECG uses 10.

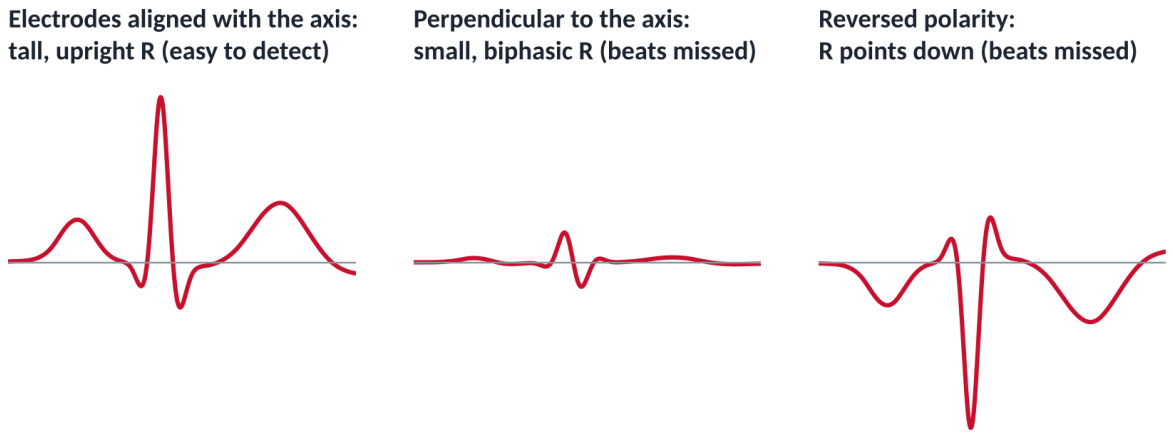


Winter, B. B., & Webster, J. G. (1983). Driven-right-leg circuit design. IEEE Trans Biomed Eng, BME-30(1), 62–66. doi:10.1109/TBME.1983.325168

Webster, J. G. (Ed.) (2010). Medical instrumentation: Application and design (4th ed.). Wiley.

The R-spike is tallest when the electrodes line up with the heart's electrical axis



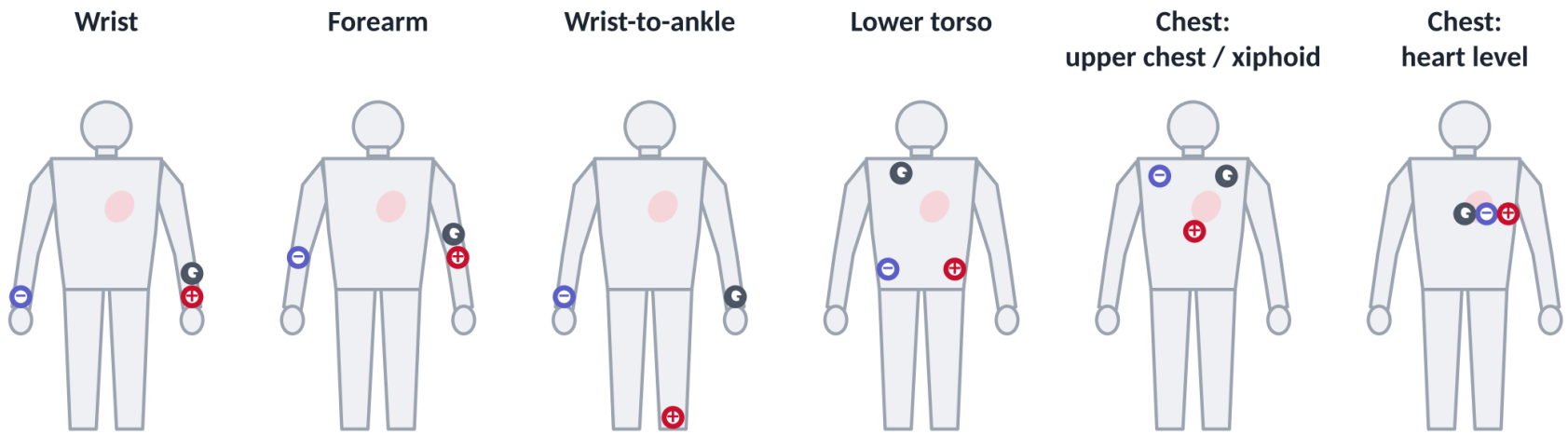


- During the QRS the heart acts like an electrical **dipole** whose mean direction usually points down and to the person's left (about +60°; normal range roughly −30° to +90°).
- Each electrode pair records the **projection** of this dipole onto the line between its electrodes (Einthoven's lead concept).
- Aligned: tall upright R. Perpendicular: small R. Reversed: R points down. Small or inverted R-spikes make detectors **miss beats** (polarity artifact, slide 53).

Malmivuo, J., & Plonsey, R. (1995). Bioelectromagnetism: Principles and applications of bioelectric and biomagnetic fields. Oxford University Press (free online at bem.fi).

Kligfield, P., et al. (2007). Recommendations for the standardization and interpretation of the electrocardiogram. Circulation, 115(10), 1306–1324. doi:10.1161/CIRCULATIONAHA.106.180200

Six common three-electrode placements trade comfort and speed against noise



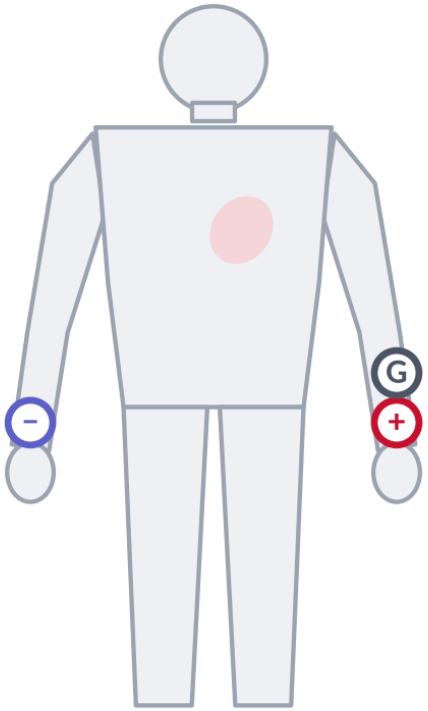
Placement	Wrist	Forearm	Wrist-to-ankle	Lower torso	Chest: upper chest / xiphoid	Chest: heart level
EMG and movement vulnerability	High	High	High	Moderate-low	High	High
Application speed	Fast	Fast	Slow	Slow	Slow	Slow
Comfort	Good	Good	Good	Poor	Poor	Poor

+ and − = active electrodes, G = reference (ground). Colour codes differ between manufacturers: always check your device manual. Figures show the participant from the front (their right is on your left).

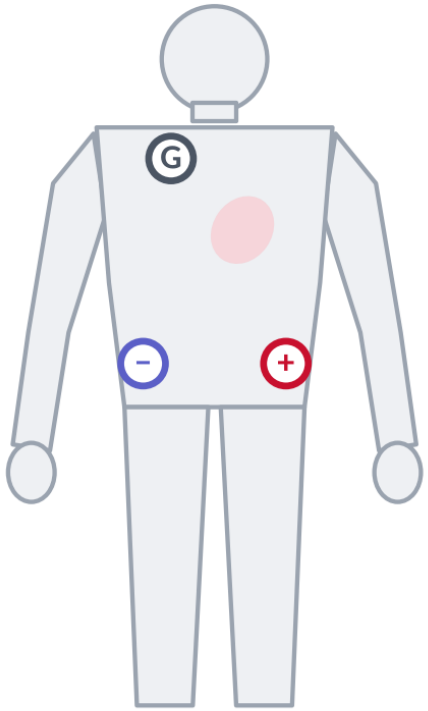
Shaffer, F., Khazan, I., & Moss, D. (2024). HRV instrumentation. BCIA HRV Biofeedback Didactic Workshop [lecture slides], practice recommendations.

Recommended: wrists for training sessions, lower torso for research

Wrist
(quick, no undressing)



Lower torso
(less muscle and movement noise)



Wrist placement

- Active (+) and reference (G) on the left wrist, active (−) on the right wrist, on the palm side.
- No undressing and quick, but arm-muscle tension and movement add noise.

Lower torso placement

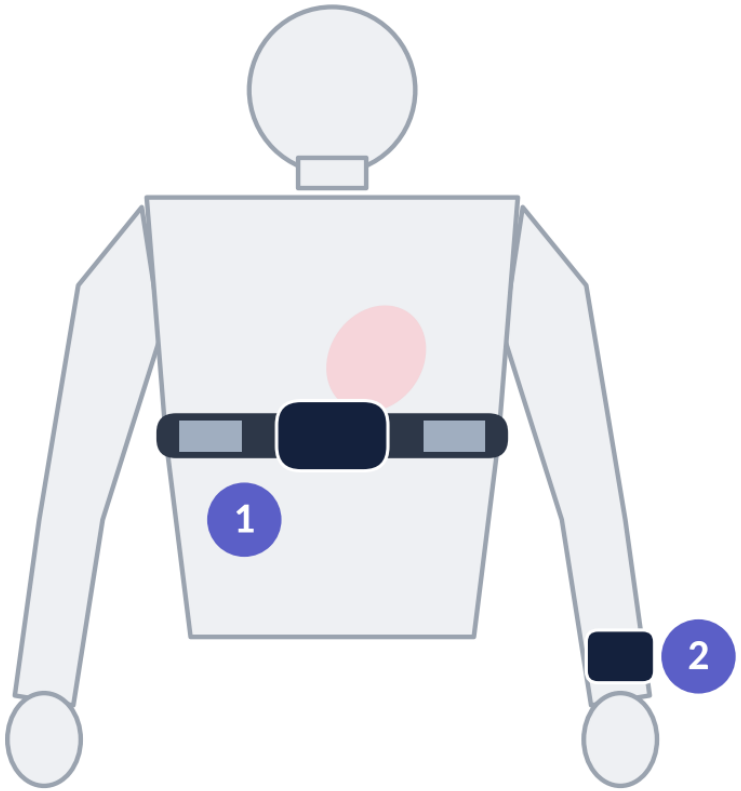
- Reference on the upper chest; actives on the left and right lower ribcage.
- Away from arm and chest muscles, so less EMG and movement noise; needs clothing moved.

Noisy at the wrists? Move to the lower torso. R-spike too small? Try the chest placements.

Shaffer, F., Khazan, I., & Moss, D. (2024). HRV instrumentation. BCIA HRV Biofeedback Didactic Workshop [lecture slides], practice recommendations.

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Chest straps bring ECG-quality intervals to everyday settings



- **1 Chest strap** (e.g. Polar H10): two dry electrodes inside the band; a small module detects R-peaks and sends RR intervals over Bluetooth Low Energy.
  - In a small validation study (10 participants), H10 RR intervals matched a Holter ECG at rest, and signal quality was even higher than the Holter during intense exercise (Gilgen-Ammann et al., 2019).
  - Moisten the electrodes and fit the band snugly just below the chest muscles.
- **2 Watches:** optical PPG on the wrist for continuous pulse rate; some add a **single-lead ECG** recorded when the user touches the watch with the other hand (spot checks, not continuous).
- **For developers:** the standard Bluetooth Heart Rate Service carries RR intervals (in units of 1/1024 s), so you can stream them into your own Python app.

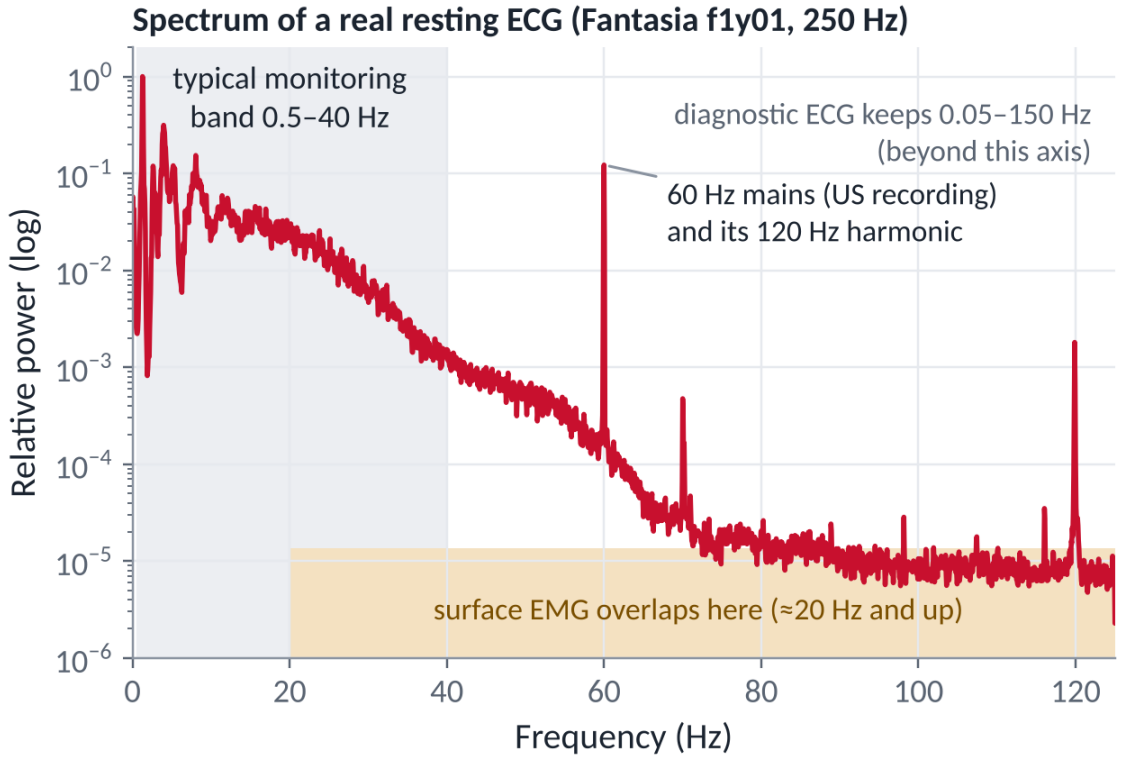
Gilgen-Ammann, R., et al. (2019). RR interval signal quality of a heart rate monitor and an ECG Holter at rest and during exercise. Eur J Appl Physiol, 119(7), 1525–1532. doi:10.1007/s00421-019-04142-5

Engineering view: ECG bandwidth, filters and sampling rate

- Most ECG energy lies **below about 40 Hz**; the QRS complex carries the highest frequencies and the steepest slopes.
- HRV and monitoring front ends often use **0.5–40 Hz**; diagnostic ECG keeps **0.05–150 Hz** (AHA recommendations).
- The sampling rate sets the **timing resolution** of each R-peak:

Sampling rate	Timing step	Comment
125 Hz	8 ms	Too coarse unless interpolated
250 Hz	4 ms	Acceptable for all HRV analyses (Kwon et al., 2018)
500 Hz	2 ms	Good margin
1000 Hz	1 ms	Common in research systems

At 100 Hz, time-domain HRV was still acceptable but frequency-domain HRV was not (Kwon et al., 2018).



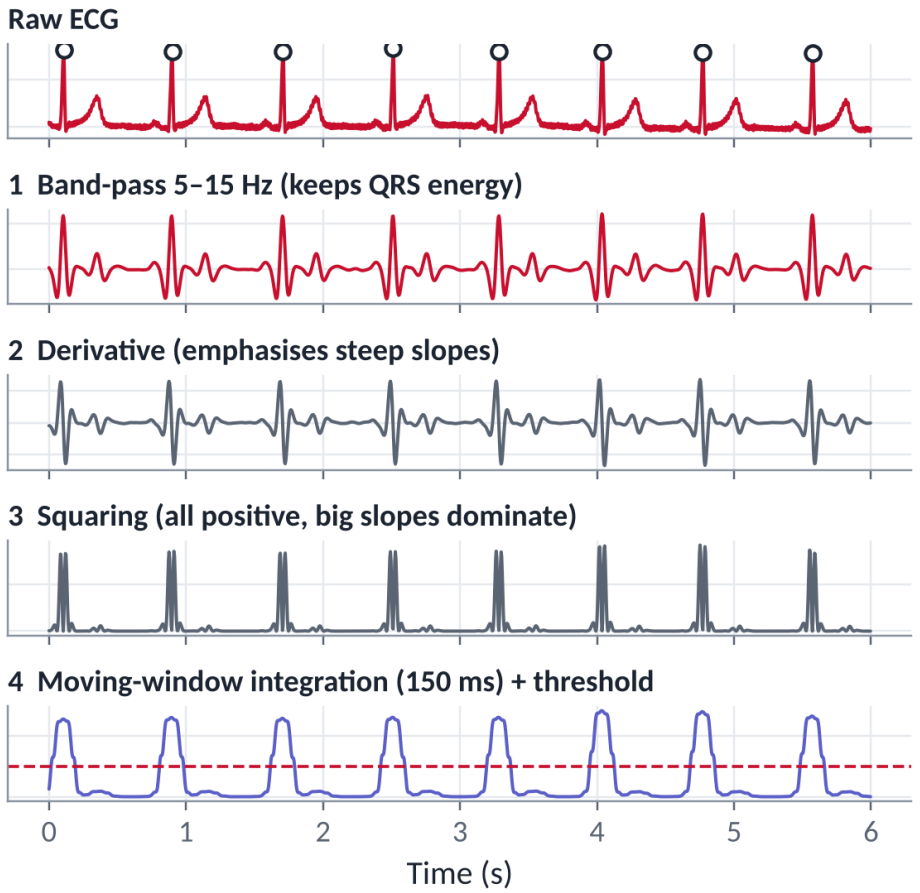
Kwon, O., et al. (2018). Electrocardiogram sampling frequency range acceptable for heart rate variability analysis. Healthc Inform Res, 24(3), 198–206. doi:10.4258/hir.2018.24.3.198

Kligfield, P., et al. (2007). Recommendations for the standardization and interpretation of the electrocardiogram. Circulation, 115(10), 1306–1324. doi:10.1161/CIRCULATIONAHA.106.180200

Data: Fantasia Database, PhysioNet (Iyengar et al., 1996; Goldberger et al., 2000).

Finding R-peaks: the Pan-Tompkins algorithm, step by step

- **1 Band-pass filter (about 5–15 Hz):** keeps the QRS energy; removes baseline wander, T waves, muscle noise and mains.
- **2 Derivative:** the QRS has the steepest slopes.
- **3 Squaring:** makes all values positive and boosts large slopes.
- **4 Moving-window integration (≈ 150 ms, one QRS width):** merges each QRS into a single hump.
- **5 Adaptive thresholds** on signal and noise peaks, a **200 ms refractory period**, and a **search-back** for missed beats.
- Designed in 1985 to run in real time on a microprocessor; still the classic baseline for new detectors.



Pan, J., & Tompkins, W. J. (1985). A real-time QRS detection algorithm. IEEE Trans Biomed Eng, BME-32(3), 230–236. doi:10.1109/TBME.1985.325532

Data: Fantasia Database, PhysioNet (Iyengar et al., 1996; Goldberger et al., 2000).

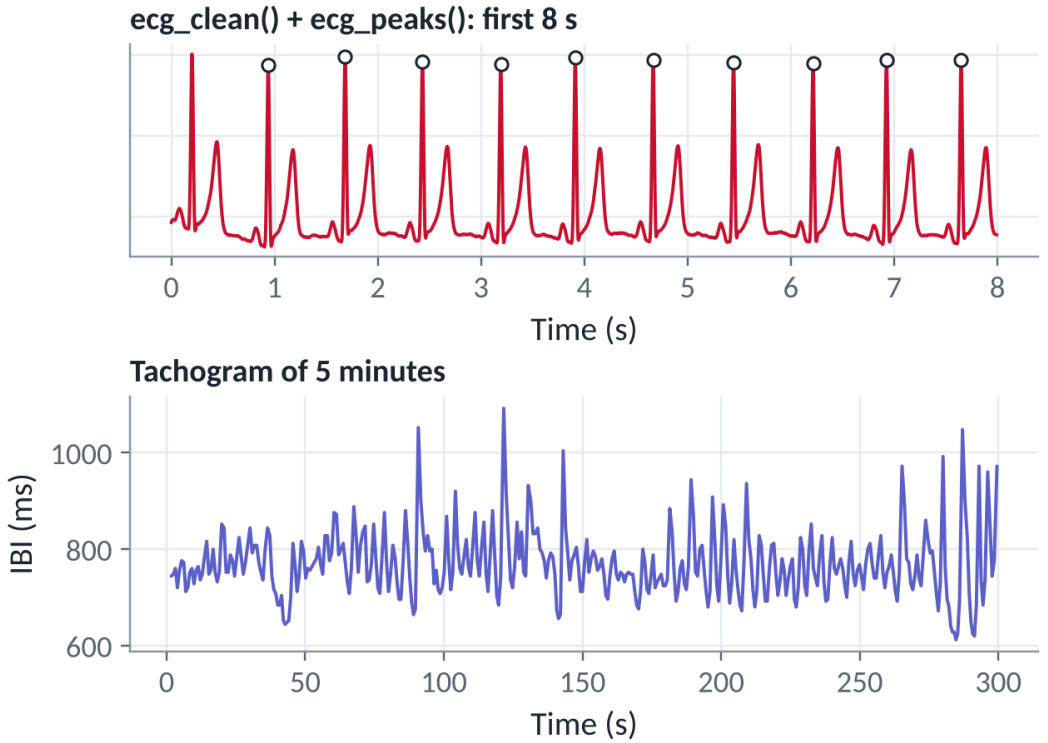
Hands-on: from an open PhysioNet recording to HRV metrics in a few lines of Python


```
import wfdb, neurokit2 as nk

# Fantasia: young adult at rest, ECG at 250 Hz
rec = wfdb.rdrecord("f1y01", pn_dir="fantasia",
                    sampto=250 * 300)  # 5 min
ecg = rec.p_signal[:, 1]   # column 1 = ECG
fs = rec.fs

clean = nk.ecg_clean(ecg, sampling_rate=fs)
_, info = nk.ecg_peaks(clean, sampling_rate=fs)
hrv = nk.hrv_time(info["ECG_R_Peaks"],
                  sampling_rate=fs)
print(hrv[["HRV_MeanNN", "HRV_SDNN",
           "HRV_RMSSD", "HRV_pNN50"]])
```

Output: MeanNN 767.5 ms · SDNN 68.2 ms · RMSSD 74.3 ms · pNN50 37.3 %



Always plot the detected peaks before trusting the numbers.

Makowski, D., et al. (2021). NeuroKit2: A Python toolbox for neurophysiological signal processing. Behav Res Methods, 53(4), 1689–1696. doi:10.3758/s13428-020-01516-y

Iyengar, N., et al. (1996). Age-related alterations in the fractal scaling of cardiac interbeat interval dynamics. Am J Physiol, 271(4), R1078–R1084. doi:10.1152/ajpregu.1996.271.4.R1078

Goldberger, A. L., et al. (2000). PhysioBank, PhysioToolkit, and PhysioNet. Circulation, 101(23), e215–e220. doi:10.1161/01.CIR.101.23.e215

ECG artifacts come either from the heart itself or from the recording

Physiological (real but not sinus rhythm)

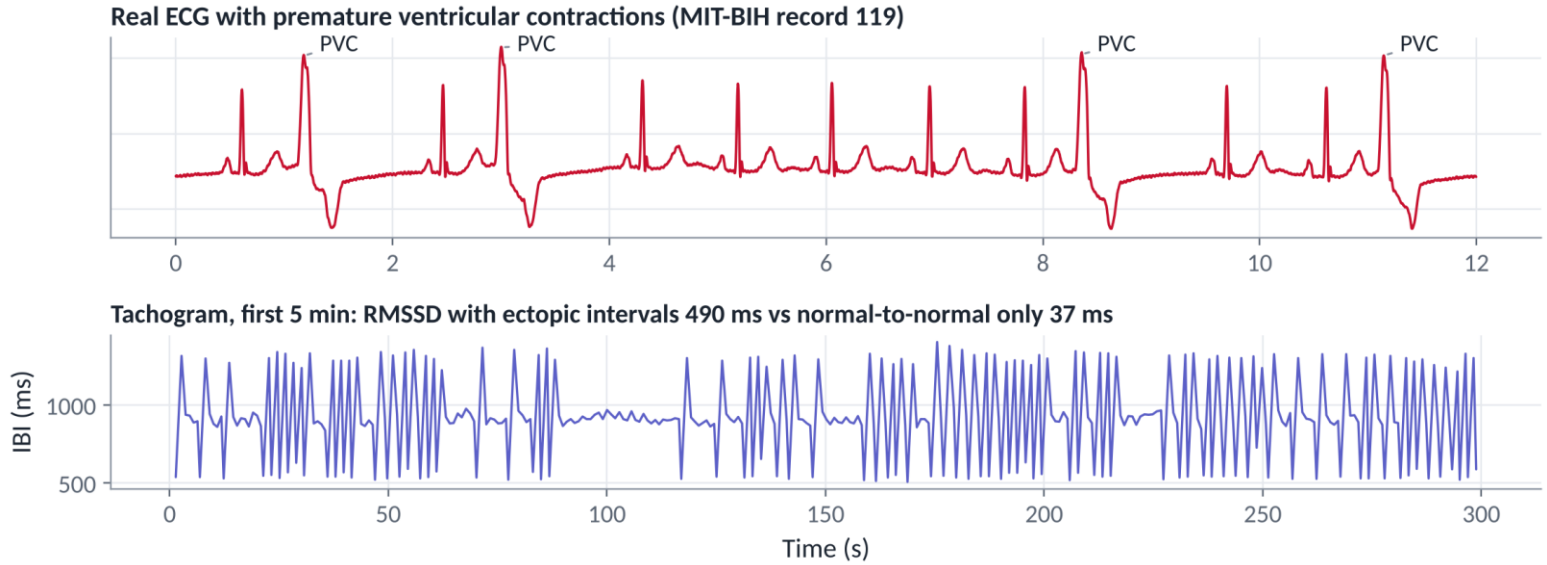
- Atrial fibrillation (AF)
- Premature atrial contractions (PACs)
- Premature ventricular contractions (PVCs)
- Effect: intervals that standard HRV analysis cannot use

Technical artifact	Typical effect on beat detection
Mains interference (50 Hz)	Fuzzy trace; extra beats if severe
Muscle (EMG)	Extra (false) beats
Movement (electrode motion)	Extra and missed beats
Respiration / baseline wander	Missed or extra beats when the drift crosses the threshold
DC offset	Jumps and drifts; extra or missed beats
Electromagnetic / radio-frequency	Bursts that look like beats
Polarity (electrode alignment)	Small or inverted R: missed beats

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Berntson, G. G., & Stowell, J. R. (1998). ECG artifacts and heart period variability: Don't miss a beat! Psychophysiology, 35(1), 127–132. doi:10.1111/1469-8986.3510127

Ectopic beats create short-long interval pairs that dominate short-term HRV



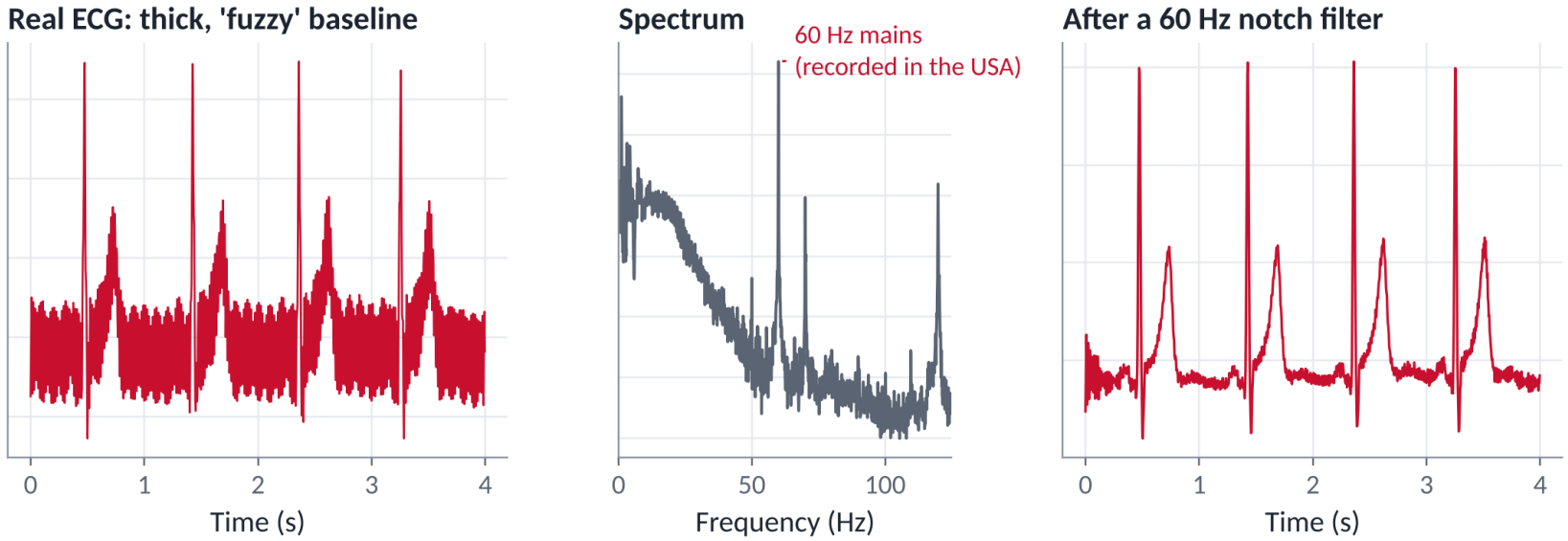
- A premature ventricular contraction (PVC) comes **early** (short interval) and is followed by a **compensatory pause** (long interval).
- In this real recording (80 PVCs among 326 beats in 5 min), RMSSD is **490 ms** with ectopic intervals included and **37 ms** using normal-to-normal intervals only.
- One artifactual interval in a 2-minute segment can noticeably distort time- and frequency-domain measures. Exclude or interpolate ectopic intervals (and report how many); if more than about 5% are affected, discard the segment. Frequent arrhythmia such as AF can make standard HRV analysis impossible.

Berntson, G. G., et al. (1997). Heart rate variability: Origins, methods, and interpretive caveats. Psychophysiology, 34(6), 623–648. doi:10.1111/j.1469-8986.1997.tb02140.x

Moody, G. B., & Mark, R. G. (2001). The impact of the MIT-BIH Arrhythmia Database. IEEE Eng Med Biol Mag, 20(3), 45–50. doi:10.1109/51.932724

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Mains interference is the most common ECG contaminant

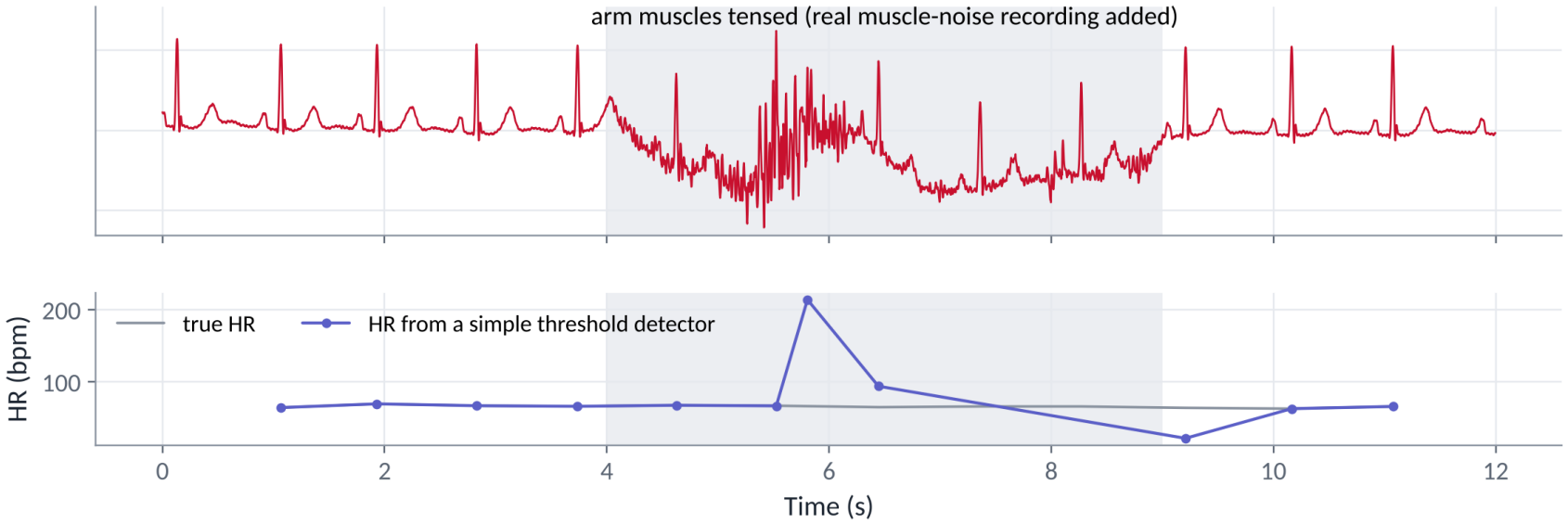


- Looks like a **thick, fuzzy baseline**. This real recording was made in the USA (60 Hz); in Europe the frequency is **50 Hz**.
- Sources: computers, monitors, fluorescent lights, power outlets and cables. ECG is far more susceptible than PPG because the electrodes pick up electric fields directly.
- **Fixes:** 50 Hz notch filter; amplifier about 1 m from other equipment; unplug unused cables; prepare the skin and use pre-gelled electrodes for **low, balanced impedance** (below 10 kΩ, within 5 kΩ); check the raw signal.

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Iyengar, N., et al. (1996). Age-related alterations in the fractal scaling of cardiac interbeat interval dynamics. Am J Physiol, 271(4), R1078–R1084. doi:10.1152/ajpregu.1996.271.4.R1078

Muscle activity (EMG) overlaps the ECG spectrum and creates false beats



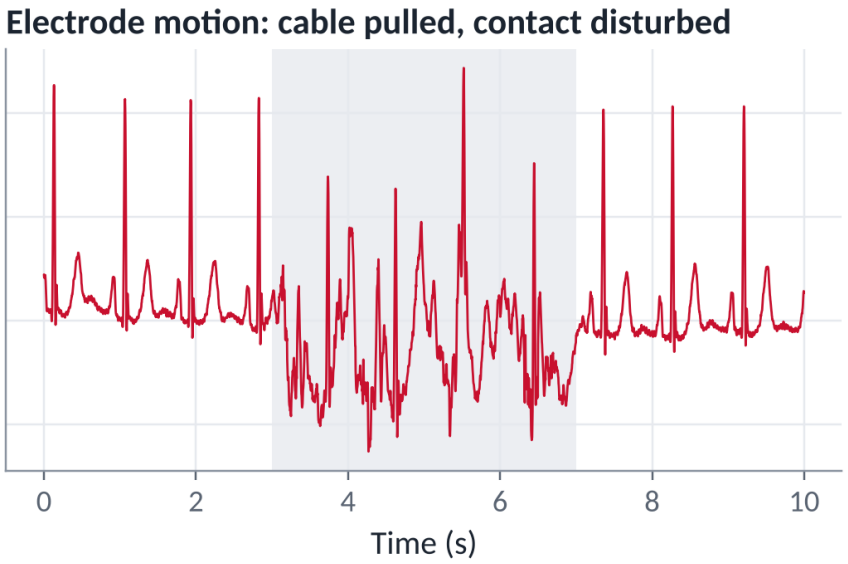
- Action potentials from skeletal muscles spread through tissue (**volume conduction**) to the ECG electrodes. Their power lies mostly between tens and a few hundred Hz, **overlapping the QRS band**, so filtering cannot separate them fully.
- Tensing the arms, gripping, shivering or talking can make a detector count spikes as beats: shorter intervals, higher HR, distorted HRV.
- **Fixes:** prefer the lower-torso placement over wrist or forearm; seat the participant comfortably with arms supported and relaxed; ask for stillness and check it; inspect the raw signal.

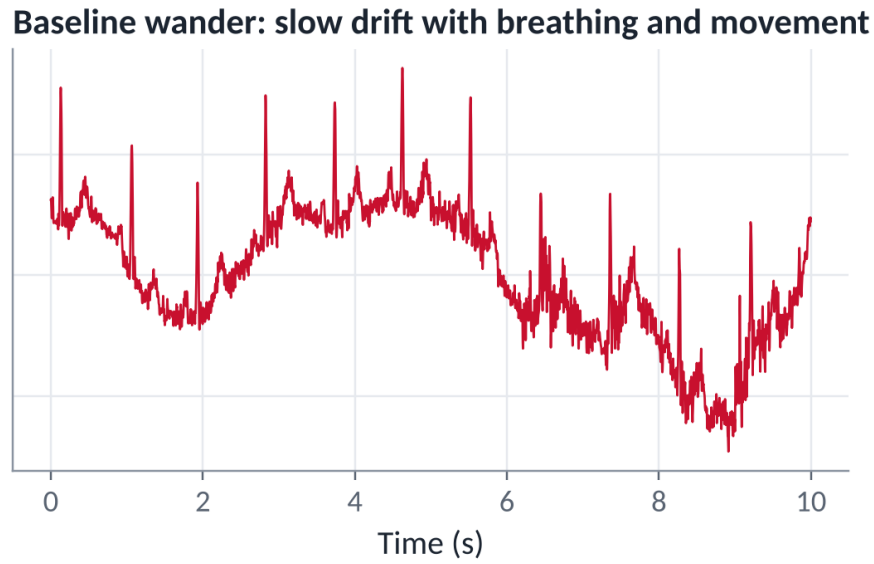
Shaffer, F., & Neblett, R. (2010). Practical anatomy and physiology: The skeletal muscle system. Biofeedback, 38(2), 47–51. doi:10.5298/1081-5937-38.2.47

Moody, G. B., Muldrow, W. E., & Mark, R. G. (1984). A noise stress test for arrhythmia detectors. Computers in Cardiology, 11, 381–384 (Noise Stress Test Database, PhysioNet).

Stern, R. M., Ray, W. J., & Quigley, K. S. (2001). Psychophysiological recording (2nd ed.). Oxford University Press.

Movement and breathing make the baseline wander and the electrodes lose contact



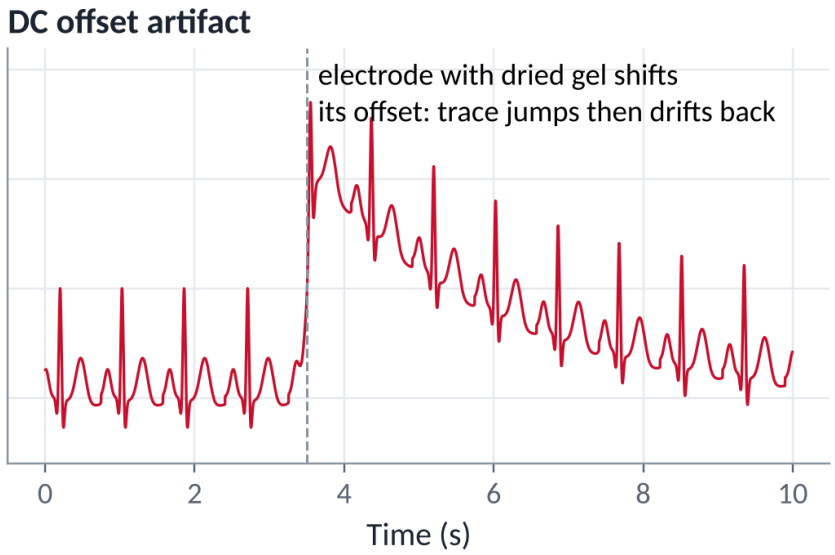


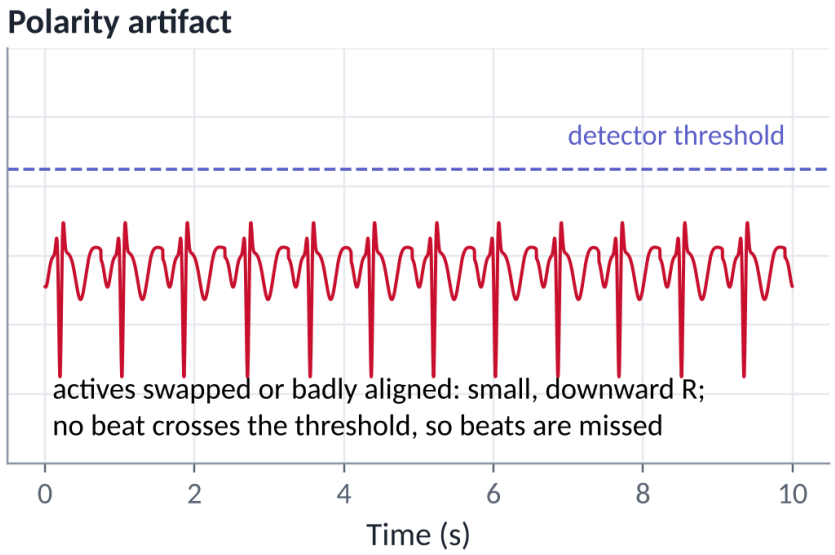
- **Electrode motion:** a pulled cable lifts the electrode partly off the skin; large swings look like extra beats.
- **Baseline wander:** slow drift with breathing or body movement, made worse by dried gel and poorly prepared skin (respiration artifact).
- **Fixes:** tape leads to clothing for strain relief; lower-torso placement; fresh pre-gelled electrodes; clean (and if needed gently abrade) the skin; relaxed seated position. In software, a high-pass filter around 0.5 Hz removes slow drift.

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Moody, G. B., Muldrow, W. E., & Mark, R. G. (1984). A noise stress test for arrhythmia detectors. Computers in Cardiology, 11, 381–384 (Noise Stress Test Database, PhysioNet).

Unbalanced electrodes and misaligned actives cause drifts and missed beats





DC offset artifact

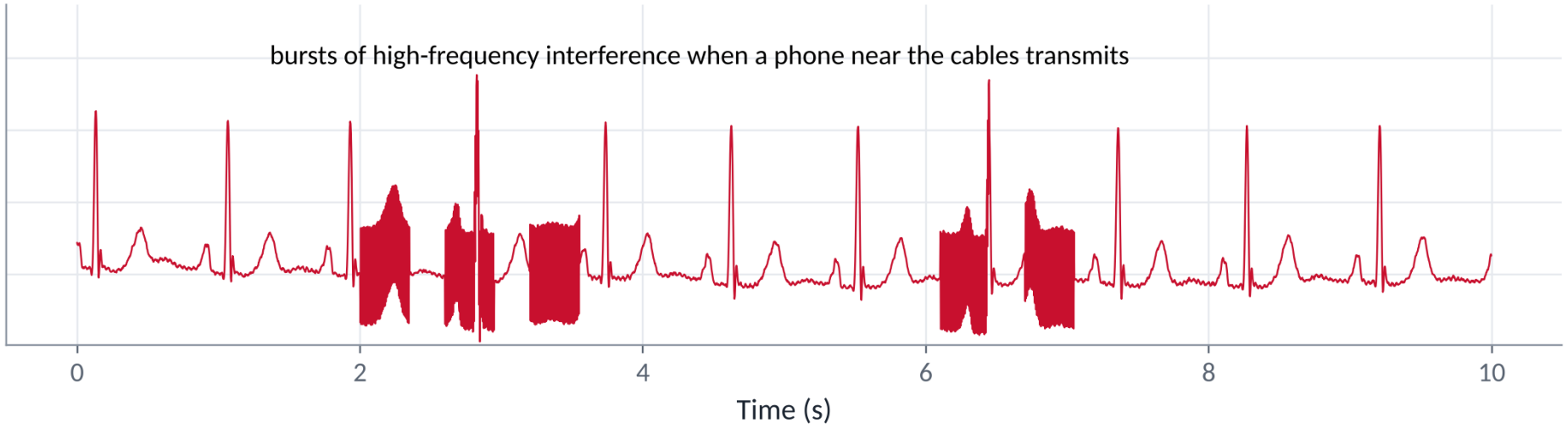
- Electrodes with different contact quality (dried gel, poor skin contact) produce different DC voltages; the trace jumps or drifts, causing extra or missed beats.
- **Fix:** clean skin, fresh electrodes with enough gel, check impedance.

Polarity artifact

- Actives swapped or perpendicular to the axis: R is small or points down, so beats are missed (long intervals).
- **Fix:** negative electrode up and to the right, positive low on the left; shift sideways until R is tall; some software can invert the signal.

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Phones, monitors and heavy equipment radiate interference into the cables



- A **mobile phone** close to sensors can produce bursts of interference: Lin & Peper (2009) recorded it in EMG sensors with the phone up to about 1.75 m away. Keep phones well away from all sensors and amplifiers.
- **Monitors and TVs** radiate electromagnetic (radio-frequency, RF) energy from the screen; so do audiovisual systems, lifts, centrifuges and X-ray machines.
- **Fixes:** phones switched off or out of the room; amplifiers behind or beside the monitor, not in front of it; participant at least 60–90 cm from the screen; keep away from high-voltage equipment.

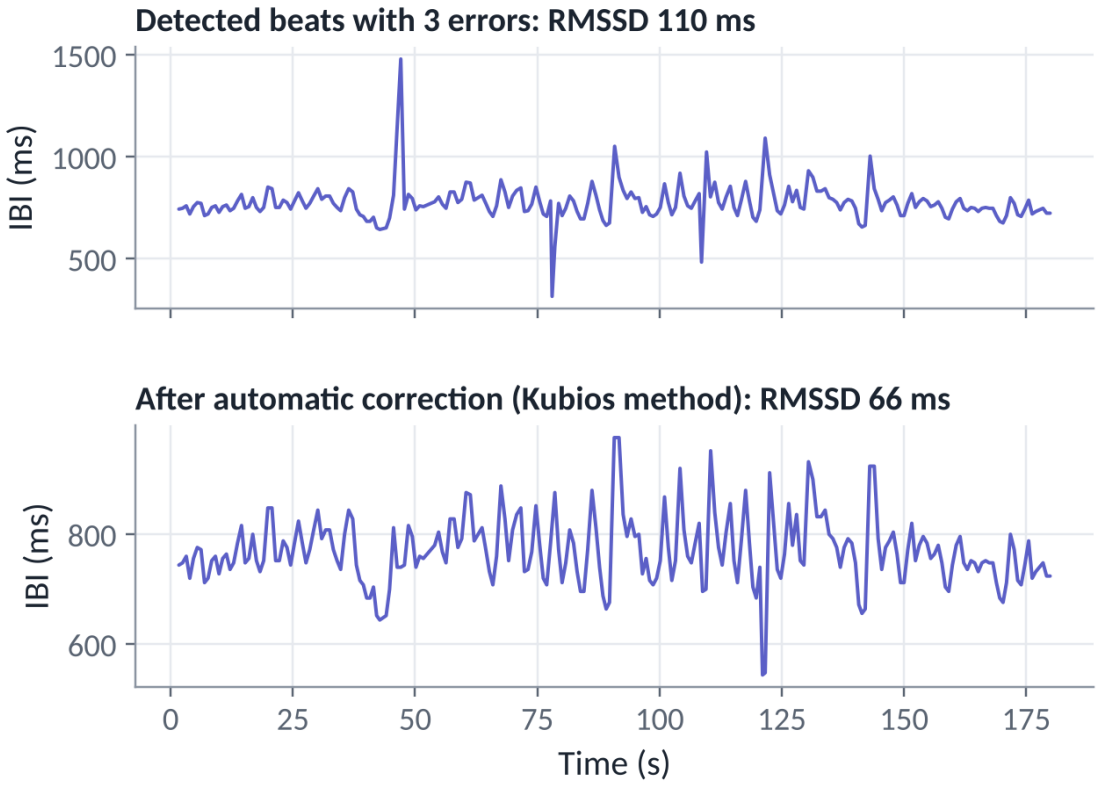
Lin, I.-M., & Peper, E. (2009). Keep cell phones and PDAs away from EMG sensors and the human body to prevent electromagnetic interference artifacts and cancer. Biofeedback, 37(3), 114–116. doi:10.5298/1081-5937-37.3.114

Montgomery, D. (2004). Introduction to biofeedback. Module 3: Psychophysiological recording. AAPB.

Software can find and correct bad intervals, but a human must check the result

Detect beats ▸ Build interval series ▸ Flag suspicious intervals ▸ Correct ▸ Review visually ▸ Analyse or discard

- **Flag:** compare each interval and successive difference with a local median; classify as ectopic, long, short, missed or extra (Lipponen & Tarvainen, 2019).
- **Correct:** insert missed beats, delete extra beats, interpolate ectopic intervals.
- **Example:** three errors raised RMSSD from 67 to 110 ms; automatic correction brought it back to 66 ms.
- **But:** on a clean real recording the same method "corrected" a genuine long pause after a deep breath and inserted two false beats. Always review corrected segments and **report how many beats were corrected**.



Lipponen, J. A., & Tarvainen, M. P. (2019). A robust algorithm for heart rate variability time series artefact correction using novel beat classification. J Med Eng Technol, 43(3), 173–181. doi:10.1080/03091902.2019.1640306

Tarvainen, M. P., et al. (2014). Kubios HRV: Heart rate variability analysis software. Comput Methods Programs Biomed, 113(1), 210–220. doi:10.1016/j.cmpb.2013.07.024

Makowski, D., et al. (2021). NeuroKit2: A Python toolbox for neurophysiological signal processing. Behav Res Methods, 53(4), 1689–1696. doi:10.3758/s13428-020-01516-y

ECG artifacts at a glance, plus the tracking test

Artifact	Looks like	Prevention or fix
Mains (50 Hz)	Thick, fuzzy baseline	Notch filter; distance; skin prep; balanced impedance
Muscle (EMG)	Irregular fast spikes when muscles tense	Lower-torso placement; relaxed, supported posture
Movement	Large swings, false beats	Strain relief; fresh electrodes; stillness
Baseline wander	Slow drift with breathing	Skin prep; fresh gel; 0.5 Hz high-pass
DC offset	Sudden jump then drift	Clean skin; electrodes with enough gel
EMI / RF	Bursts of high-frequency noise	Phones off; distance from monitors and equipment
Polarity	Small or inverted R; missed beats	Align actives with the heart's axis
Ectopic / AF	Short-long pairs; irregular rhythm	Exclude or interpolate; discard if more than ~5%

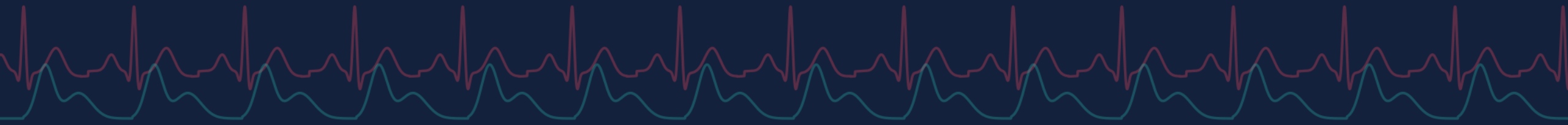
Tracking test: ask the participant to inhale slowly, then exhale. Instantaneous HR should rise during inhalation and fall during exhalation. If it does not, check the signal before you start training.

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

PART 3

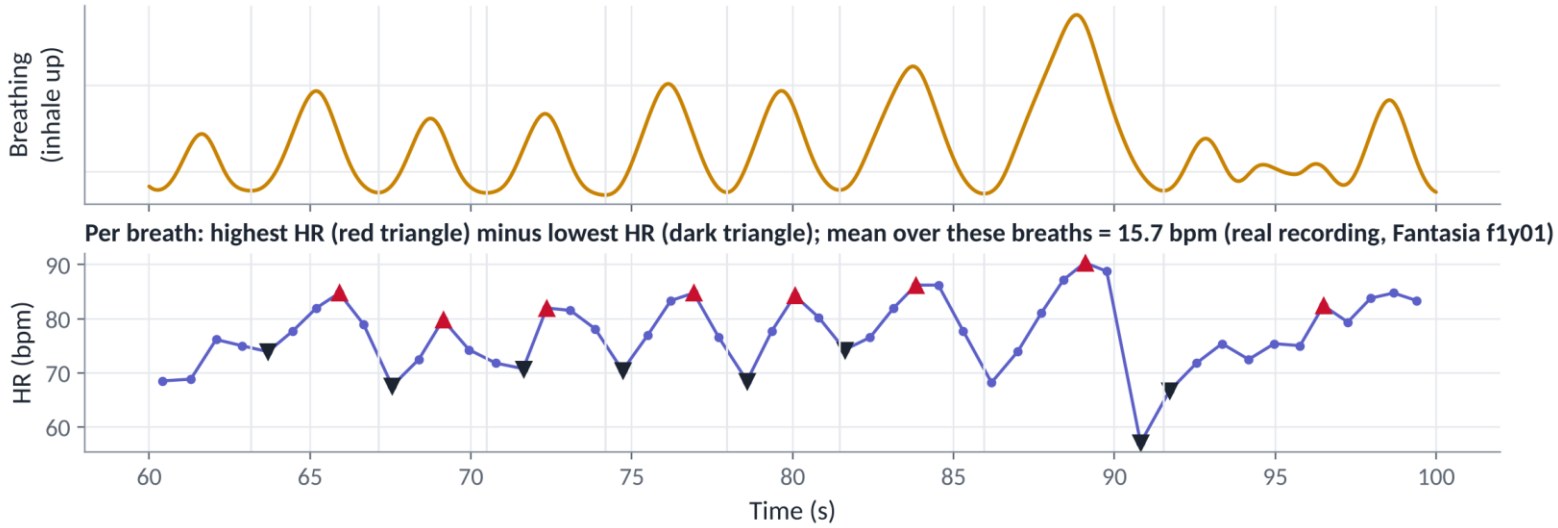
Measuring breathing

Why HRV work needs a respiration signal, which sensors exist, and how to fit and check a respiration belt



Breathing is both a source of HRV and the main training target in HRV biofeedback

- RSA links each breath to a heart-rate oscillation, so interpreting HRV requires knowing **how fast and how deeply** the person breathes. Slow breathing moves RSA from the HF into the LF band.
- HRV biofeedback guides breathing towards the **resonance frequency**, about 4.5 to 6.5 breaths/min, determined individually.
- **HR Max – HR Min:** for each breath, highest minus lowest heart rate, averaged over breaths. It needs a breathing signal to define each breath.



Lehrer, P. M., et al. (2000). Resonant frequency biofeedback training to increase cardiac variability: Rationale and manual for training. Appl Psychophysiol Biofeedback, 25(3), 177–191. doi:10.1023/A:1009554825745

Lehrer, P. M., & Gevirtz, R. (2014). Heart rate variability biofeedback: How and why does it work? Front Psychol, 5, 756. doi:10.3389/fpsyg.2014.00756

Data: Fantasia Database, PhysioNet (Iyengar et al., 1996; Goldberger et al., 2000).

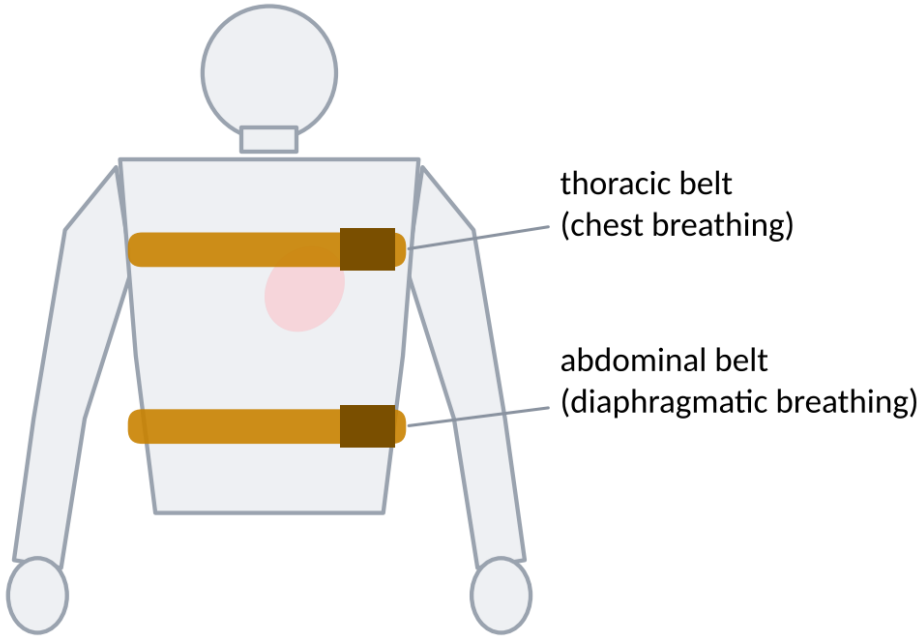
Many ways to sense breathing; biofeedback mostly uses a stretch belt

Method	Principle	Strengths	Limitations
Strain-gauge belt (respirometer)	Band stretch changes electrical resistance	Simple, cheap, standard in biofeedback	Relative units; slips; depends on fit
Piezoelectric belt	Stretch generates electrical charge	Robust, low power	Responds to change; poor for very slow or static signals
Respiratory inductance plethysmography (RIP)	Coil in the belt; inductance follows the cross-section	Good volume estimate after calibration; sleep labs	More expensive
Nasal thermistor or airflow	Temperature or pressure of breath at the nose	Direct airflow; detects pauses	Uncomfortable for long sessions
Capnometer (end-tidal CO_2)	Infrared CO_2 measurement in exhaled air	Shows over-breathing (low CO_2)	Costly; nasal cannula
Derived from ECG or PPG	Breathing modulates amplitude, baseline and rate	No extra sensor	Less reliable; mainly research

Massaroni, C., et al. (2019). Contact-based methods for measuring respiratory rate. Sensors, 19(4), 908. doi:10.3390/s19040908

Charlton, P. H., et al. (2018). Breathing rate estimation from the electrocardiogram and photoplethysmogram: A review. IEEE Rev Biomed Eng, 11, 2–20. doi:10.1109/RBME.2017.2763681

Fit the belt at the end of a full exhalation: snug but comfortable

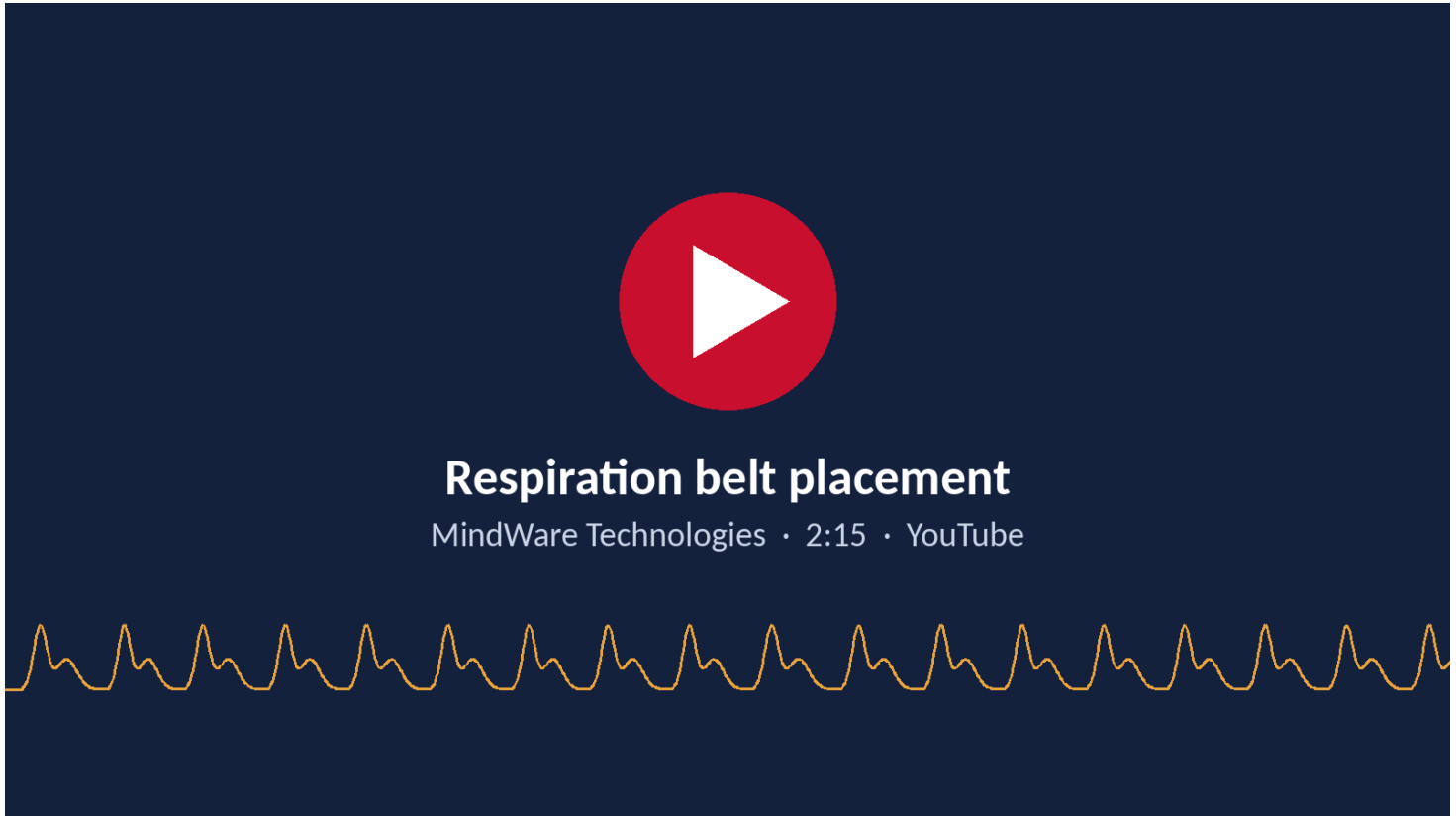


dark block = sensing element (stretch, piezo or inductive)

1. **Choose the position:** abdomen (around the navel) for diaphragmatic breathing; add a chest belt to compare chest and abdominal breathing.
2. Ask the participant to **breathe out completely**.
3. During the pause after exhalation, **tighten the band slightly**: it must stay stretched even at full exhalation.
4. **Check the trace:** smooth waves, no flat tops or bottoms, enough amplitude to find peaks and troughs.
5. **Re-check after posture changes:** belts slip.

Shaffer, F., Khazan, I., & Moss, D. (2024). HRV instrumentation. BCIA HRV Biofeedback Didactic Workshop [lecture slides], practice recommendations.
Massaroni, C., et al. (2019). Contact-based methods for measuring respiratory rate. Sensors, 19(4), 908. doi:10.3390/s19040908

Video: placing a respiration belt correctly (2 min)



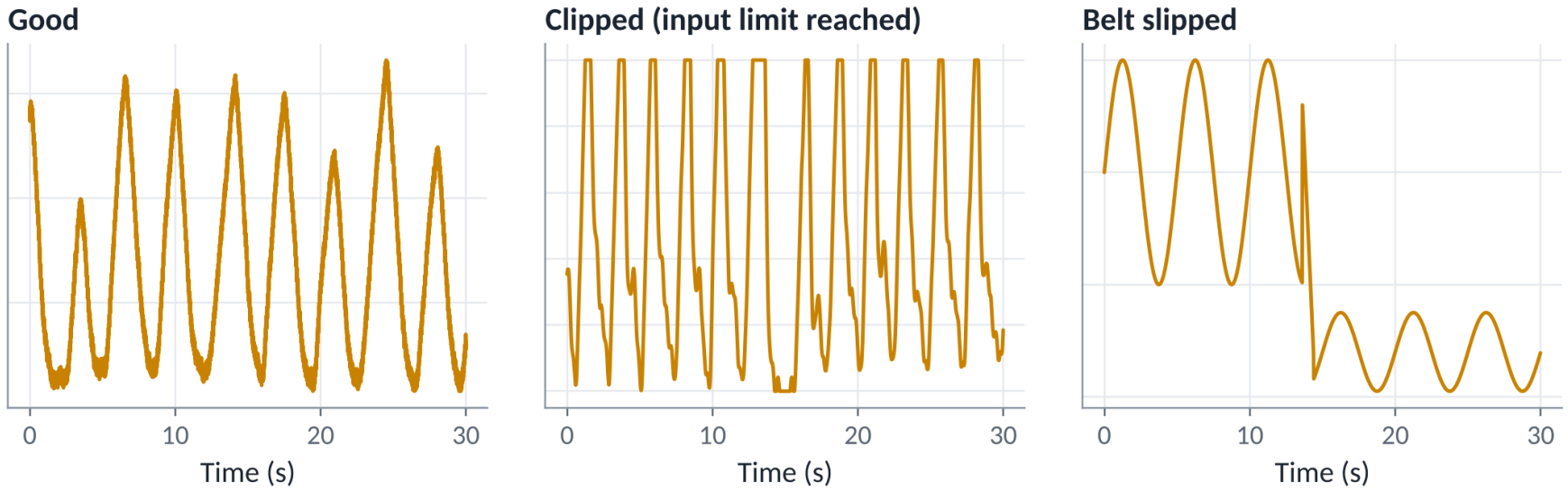
If the video does not play here, open it on YouTube

While you watch, look for:

- Where the belt sits on the body.
- How tight it should be, and when to tighten it.
- How to check the signal once the belt is on.

Video: MindWare Technologies. Respiration belt placement training. YouTube. https://www.youtube.com/watch?v=eDIJ7AiKu8s

A good respiration trace has clear peaks and troughs, without clipping or jumps



- Values are **relative (arbitrary units)**, not litres: amplitude depends on belt position, girth, clothing and tension. Compare rate and pattern across sessions; compare amplitude only within a session.
- **Clipping:** the signal hits the input range (flat tops). Re-fit or reduce the gain.
- **Loose or slipping belt:** sudden offset jump and small amplitude. Re-fit (slide 60).

Shaffer, F., Khazan, I., & Moss, D. (2024). HRV instrumentation. BCIA HRV Biofeedback Didactic Workshop [lecture slides], practice recommendations.

Data: Fantasia Database, PhysioNet (Iyengar et al., 1996; Goldberger et al., 2000).

Data: BIDMC PPG and Respiration Dataset, PhysioNet (Pimentel et al., 2017; Goldberger et al., 2000).

Quick tracking tests confirm the breathing display tells the truth

Movement tracking

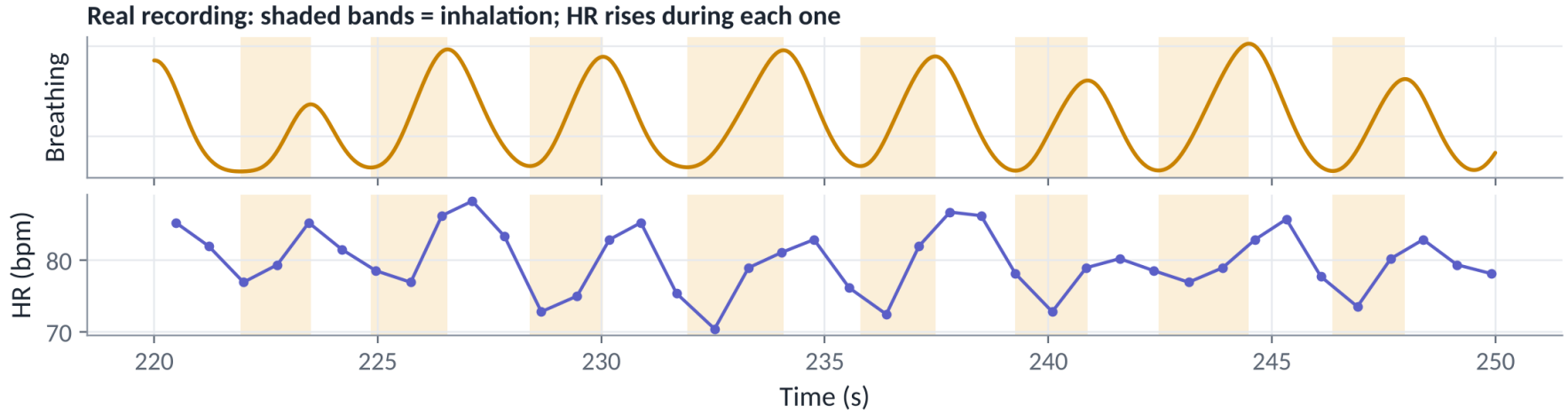
Ask the participant to inhale slowly, then exhale. The trace must rise and fall exactly with the abdomen: watch the body and the screen together.

Rate check

Count the breaths yourself for 60 s and compare with the displayed respiration rate.

Heart-breath check

Heart rate should rise during inhalation and fall during exhalation (RSA, lower panel).

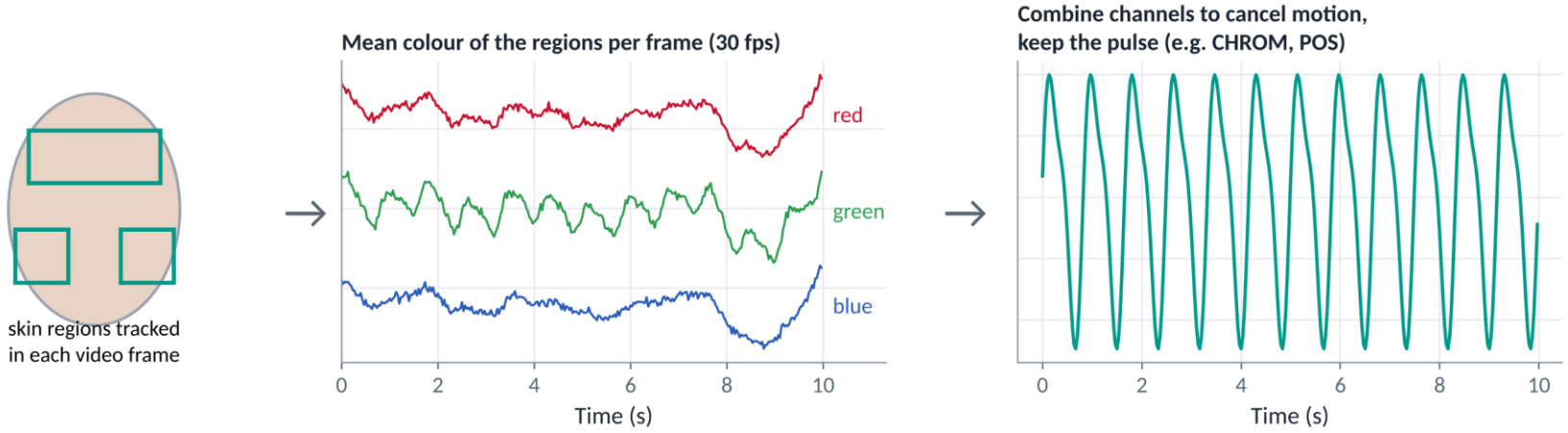


Shaffer, F., Khazan, I., & Moss, D. (2024). HRV instrumentation. BCIA HRV Biofeedback Didactic Workshop [lecture slides], practice recommendations.

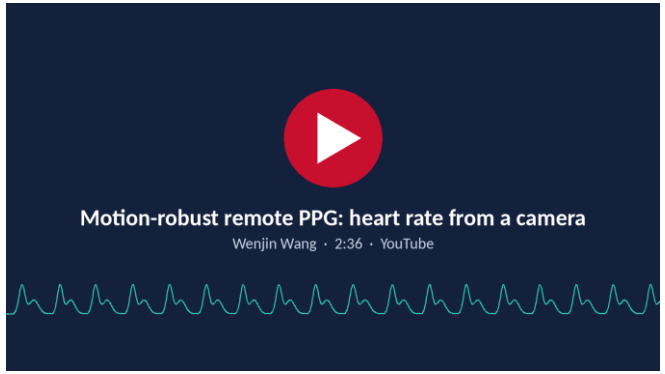
Data: Fantasia Database, PhysioNet (Iyengar et al., 1996; Goldberger et al., 2000).

Beyond contact sensors: a camera can measure the pulse from skin colour

IN PRACTICE



- **Remote PPG (rPPG):** an ordinary camera records tiny colour changes of the skin with each beat (Verkruysse et al., 2008).
- The green channel carries most pulse information. Algorithms such as **CHROM** (de Haan & Jeanne, 2013) and **POS** (Wang et al., 2017) combine R, G and B to cancel motion and lighting changes; deep learning is increasingly used.
- Challenges: motion, lighting, skin tone, video compression, and **privacy** of face video.



Open on YouTube (2.5 min)

Verkruysse, W., et al. (2008). Remote plethysmographic imaging using ambient light. Opt Express, 16(26), 21434–21445. doi:10.1364/OE.16.021434

de Haan, G., & Jeanne, V. (2013). Robust pulse rate from chrominance-based rPPG. IEEE Trans Biomed Eng, 60(10), 2878–2886. doi:10.1109/TBME.2013.2266196

Wang, W., et al. (2017). Algorithmic principles of remote PPG. IEEE Trans Biomed Eng, 64(7), 1479–1491. doi:10.1109/TBME.2016.2609282

Video: Wang, W. Motion robust remote-PPG (camera-based heart-rate monitoring). YouTube. https://www.youtube.com/watch?v=Q9MK-vtWzUM

Choosing equipment: match the system to the question

IN PRACTICE

Category	Examples	Sensors	Good for
Multichannel biofeedback systems	Thought Technology ProComp, Mind Media NeXus	ECG, PPG, respiration, EMG, skin conductance, temperature	Certification-level training; research with raw signals
Dedicated HRV biofeedback	HeartMath emWave Pro, Inner Balance	PPG (ear or finger)	Coherence and resonance breathing training
Chest strap + app	Polar H10 with apps such as Elite HRV, Optimal HRV, Kubios	ECG-based RR intervals	Low-cost training and field studies
Smartwatches and rings	Many brands	PPG + accelerometer	Long-term trends and sleep; limited beat-level access

- HRV biofeedback needs at least **heart rate (ECG or PPG) + respiration**; EMG of shoulder and neck muscles and skin conductance or temperature are useful extras.
- Use **ECG** if you plan to publish. Compare **total cost**: amplifier, sensors and software licences.
- For engineers: check that the system can **export raw data** (waveforms and beat times) before buying.

Shaffer, F., Khazan, I., & Moss, D. (2024). HRV instrumentation. BCIA HRV Biofeedback Didactic Workshop [lecture slides], practice recommendations.

Quigley, K. S., et al. (2024). Publication guidelines for human heart rate and heart rate variability studies in psychophysiology, Part 1. Psychophysiology, 61(9), e14604. doi:10.1111/psyp.14604

Free tools and open data let you practise the whole pipeline

IN PRACTICE

Tool or resource	What it is	Use it for
Kubios HRV Scientific Lite (free, non-commercial) / **Scientific** (paid)	Desktop HRV analysis software	Standard HRV reports, artifact correction
NeuroKit2 (Python)	Open-source physiological signal toolbox	ECG, PPG and respiration processing; HRV
HeartPy (Python)	Heart-rate toolkit designed for noisy PPG	PPG peak detection
WFDB (Python / MATLAB)	PhysioNet data access library	Reading open datasets and annotations
PhysioNet databases used in this lecture	Open physiological recordings	MIT-BIH Arrhythmia (ectopic beats), Fantasia (rest ECG + breathing), BIDMC (ECG + PPG + breathing), Noise Stress Test (real artifacts)

Suggested exercise: take a clean Fantasia record, add Noise Stress Test noise at different levels, and measure how detection errors change RMSSD.

Tarvainen, M. P., et al. (2014). Kubios HRV: Heart rate variability analysis software. Comput Methods Programs Biomed, 113(1), 210–220. doi:10.1016/j.cmpb.2013.07.024
Makowski, D., et al. (2021). NeuroKit2: A Python toolbox for neurophysiological signal processing. Behav Res Methods, 53(4), 1689–1696. doi:10.3758/s13428-020-01516-y
van Gent, P., et al. (2019). HeartPy: A novel heart rate algorithm for the analysis of noisy signals. Transp Res F, 66, 368–378. doi:10.1016/j.trf.2019.09.015
Goldberger, A. L., et al. (2000). PhysioBank, PhysioToolkit, and PhysioNet. Circulation, 101(23), e215–e220. doi:10.1161/01.CIR.101.23.e215

Reference values help spot implausible recordings, but norms vary with age and method

IN PRACTICE

Measure	Typical adult value at rest	Source
Heart rate	60–80 beats/min (non-athletes); 40–60 (athletes)	Khazan, 2019
RMSSD	≈ 30–100 ms; short-term mean 42 ± 15 ms	Khazan, 2019; Nunan et al., 2010
SDNN	> 50 ms; short-term mean 50 ± 16 ms	Khazan, 2019; Nunan et al., 2010
ln HF (ln ms^2)	4.5–7.5	Khazan, 2019
Respiration rate	12–14 breaths/min (biofeedback norm; clinical ranges are wider)	Khazan, 2019
End-tidal CO_2	35–45 mmHg	Khazan, 2019
Oxygen saturation	95–98%	Khazan, 2019
Skin conductance level	≤ 5 µS	Khazan, 2019
Finger temperature	31–35 °C	Khazan, 2019



HRV falls with age and differs by sex, fitness, posture and recording length. Use norms to flag implausible values, not to diagnose.

Khazan, I. (2019). A guide to normal values for biofeedback. In D. Moss & F. Shaffer (Eds.), Physiological recording technology and applications in biofeedback and neurofeedback (pp. 2–6). AAPB.

Nunan, D., et al. (2010). A quantitative systematic review of normal values for short-term heart rate variability in healthy adults. Pacing Clin Electrophysiol, 33(11), 1407–1417. doi:10.1111/j.1540-8159.2010.02841.x

Shaffer, F., & Ginsberg, J. P. (2017). An overview of heart rate variability metrics and norms. Front Public Health, 5, 258. doi:10.3389/fpubh.2017.00258

Medications and everyday substances shift HR and HRV: ask about them and record them

IN PRACTICE

Usually raise HR and/or lower HRV

- Tricyclic antidepressants
- Stimulants (amphetamines, methylphenidate, cocaine)
- Nicotine
- Alcohol (acute)
- Beta-2 agonist asthma inhalers (raise HR)

Usually lower HR and raise HRV

- Beta-blockers

Small or inconsistent effects

- SSRIs (no significant HRV change in a meta-analysis)
- Caffeine (mixed findings: standardise intake)
- NSAIDs

In practice: before each session ask about medication, caffeine, nicotine, alcohol and recent exercise; record at consistent times of day; report these factors in research.

Kemp, A. H., et al. (2010). Impact of depression and antidepressant treatment on heart rate variability: A review and meta-analysis. Biol Psychiatry, 67(11), 1067–1074. doi:10.1016/j.biopsych.2009.12.012

Quigley, K. S., et al. (2024). Publication guidelines for human heart rate and heart rate variability studies in psychophysiology, Part 1. Psychophysiology, 61(9), e14604. doi:10.1111/psyp.14604

Sammito, S., et al. (2024). Guideline for the application of heart rate and heart rate variability in occupational medicine and occupational health science. J Occup Med Toxicol, 19(1), 15. doi:10.1186/s12995-024-00414-9

Sensors touch skin: clean, disinfect and use disposables

IN PRACTICE

Hands

Participant and operator wash their hands before the session.

Surfaces

Disinfect chair arms and headrests with wipes rated against bacteria and viruses; weak alcohol solutions are not enough.

Reusable sensors

Clean and disinfect sensors and belts after each participant, following the manufacturer's instructions (some agents damage sensor materials).

Disposables

Single-use pre-gelled ECG electrodes; never reuse them.

Hagedorn, D. (2014). Infection risk mitigation for biofeedback providers. Biofeedback, 42(3), 93–95. doi:10.5298/1081-5937-42.3.06

Moss, D., et al. (2019). Care for biofeedback and neurofeedback instrumentation. Biofeedback, 47(1), 12–21. doi:10.5298/1081-5937-47.1.04

Recording-session checklist

IN PRACTICE

Before

- Room at least 23 °C and quiet
- Phones off; amplifier 1 m from other electronics
- 50 Hz notch filter on
- Ask about medication, caffeine, nicotine, alcohol, exercise

Set-up

- ECG: clean skin, fresh electrodes, impedance below 10 kΩ and balanced, leads taped
- PPG: pad side, snug, covered, hand at heart level
- Belt: fitted after a full exhalation

Check

- Tracking tests: HR with breathing, belt with abdomen, breath count
- Inspect the raw signals
- Allow a few minutes of rest before the baseline recording

After

- Inspect beat detection
- Correct or flag artifacts; discard segments with more than about 5% bad intervals
- Document every correction

Shaffer, F., & Combatalade, D. C. (2013). Don't add or miss a beat: A guide to cleaner heart rate variability recordings. Biofeedback, 41(3), 121–130. doi:10.5298/1081-5937-41.3.04

Laborde, S., et al. (2017). Heart rate variability and cardiac vagal tone in psychophysiological research: Recommendations for experiment planning, data analysis, and data reporting. Front Psychol, 8, 213. doi:10.3389/fpsyg.2017.00213

Quigley, K. S., et al. (2024). Publication guidelines for human heart rate and heart rate variability studies in psychophysiology, Part 1. Psychophysiology, 61(9), e14604. doi:10.1111/psyp.14604

Key takeaways

IN PRACTICE

1. **Every HRV value depends on the whole chain:** physiology, sensor, electronics, detection and cleaning.
2. **ECG gives the most precise beat timing:** PPG is convenient, but its intervals include pulse travel time.
3. **Know the typical failure modes:** PPG: movement, light, pressure, cold. ECG: mains, muscle, contact, polarity.
4. **Missed and extra beats inflate HRV:** one bad interval can distort a 2-minute segment.
5. **Record breathing:** it drives HRV and is the target of HRV biofeedback.
6. **Prevent, inspect, correct, document:** and never trust a number you have not checked against the raw signal.

Optional at home: BPM biosignals. ECG: Common artefacts and how to avoid them [video]. YouTube. https://www.youtube.com/watch?v=VsZLtW5aSfU

Optional at home: Mind Drip. Everything you should know about heart rate variability (HRV) [video]. YouTube. https://www.youtube.com/watch?v=zUyuUoU7lAQ

Glossary (1 of 2)

Term	Meaning
Arteriole	Small muscular artery whose diameter controls local blood flow
Artifact	Any part of a signal not caused by the event being measured; in HRV, errors in interbeat intervals
Atrial fibrillation (AF)	Disorganised electrical activity of the atria, causing an irregular heartbeat
Blood volume pulse (BVP)	Phasic change in tissue blood volume with each heartbeat, measured by PPG
CMRR	Common-mode rejection ratio: how well a differential amplifier suppresses signals common to both inputs
DC offset artifact	Jumps or drifts caused by unequal electrode-skin potentials
Dicrotic notch	Small dip in the PPG pulse between the direct and the reflected wave
Ectopic beat (PAC, PVC)	Premature beat starting outside the SA node, in the atria (PAC) or ventricles (PVC)
EMG artifact	Electrical activity of skeletal muscles picked up by ECG electrodes
HR Max – HR Min	Mean, over breaths, of the highest minus the lowest heart rate within each breath
IBI, RR, NN interval	Time between successive beats; RR from ECG R-peaks; NN after removing artifacts and ectopic beats

Glossary (2 of 2)

IN PRACTICE

Term	Meaning
Mains (line) interference	50 Hz (Europe) or 60 Hz (Americas) noise from the power supply and nearby equipment
Notch filter	Filter that removes a narrow frequency band, such as 50 Hz
Photoplethysmography (PPG)	Optical measurement of blood volume changes using an LED and a photodetector
Polarity artifact	Small or inverted R-spikes caused by electrodes misaligned with the heart's electrical axis
Pulse amplitude variability (PAV)	Beat-to-beat variation in PPG pulse amplitude
Pulse arrival time (PAT)	Delay from the ECG R-peak to the arrival of the pulse at the PPG site
Resonance frequency breathing	Breathing at the individual rate (about 4.5–6.5 breaths/min) that maximises heart-rate oscillations
Respiratory sinus arrhythmia (RSA)	Heart rate rising during inhalation and falling during exhalation
Respirometer	Belt sensor that measures chest or abdominal expansion
RMSSD, SDNN	Root mean square of successive differences; standard deviation of NN intervals
Tachogram	Plot or series of interbeat intervals against time
Tracking test	Quick check that the display follows the participant's physiology (e.g. HR rising with inhalation)